



A taxonomy of machine learning techniques for construction cost estimation

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Abstract

Construction projects require significant funding and are exposed to several risks. Public construction projects require a major proportion of the annual government budget. Their actual cost estimation concerns a known and existing problem for the construction sector, while several project failures in terms of budget extension can be documented around the world. Accurate construction cost predictions are essential in mitigating time-related risks and play a crucial role in the decision-making process for managers. Inaccurate cost estimations can result in investment project disruptions. Research about machine learning (ML) techniques regarding construction cost estimation is intensifying, which aims to develop new ML techniques or update existing ones. This article contains a systematic literature review of ML techniques for construction project cost estimation. This review included an in-depth analysis of 219 studies, which contain the most prominent machine learning techniques. This article attempts to define a classification of the identified ML techniques, with the following criteria: intelligent technique that was followed and the application domain. The taxonomy that was generated contains ML techniques about construction cost estimation and their application, which offers useful guidance for both researchers and practitioners.

Keywords Construction projects · Cost estimation · Machine learning · Systematic literature review · Taxonomy

Abbreviations

AHP	Analytic hierarchy process	CI	Clustering improvement
AI	Artificial intelligence	CNN	Convolutional neural network
AIM	Attribute impact	CONP	Construction projects
AMC	Area-machine-correlation	CRA	Crossvalidation regression analysis
ANFIS	Adaptive network-based fuzzy inference system	DBML	Deep Boltzman machine learning
ANN	Artificial neural network	DBN	Deep belief network
BN	Bayesian network	DF	Decision forest
BDT	Boosted decision tree	DIR	Data inconsistency rate
BO	Bayesian optimization	DM	Deterministic method
BOD	Biochemical oxygen demand	DNN	Deep neural network
BPNN	Back propagation neural network	DT	Decision tree
BR	Bridges	EGBS	Extreme gradient boosting system
BRT	Boosted regression trees	ET	Extra trees
BS	Building structures	FANN	Fast artificial neural network
CBR	Case based reasoning	FL	Fuzzy logic
CHNN	Chaotic neural network	FLRA	Fuzzy linear regression analysis
		FM	Fuzzy mathematics
		FNN	Fuzzy neural network
		GA	Genetic algorithm
		GBT	Gradient boosted trees
		GD	Gradient descent
		GEP	Gene expression programming
		GFM	Grey forecasting model
		GLM	Generalized Linear models

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GNFM	Genetic fuzzy model
GPR	Gaussian process regression
GRNN	General regression neural network
HR	Highways/roadways
ICA	Independent component analysis
IEEE	Institute of electrical and electronics engineers
LAR	Least angle regression
LEED	Leadership in energy and environmental design
LR	Linear regression
LS-SVM	Least squares support vector machine
LSTMNN	Long short-term memory neural network
MBFP	Military barrack and facility projects
MCS	Monte Carlo simulation
ML	Machine learning
MLFFN	Multi-layer feed forward network
MLP	Multi-layer perceptron
MPSO	Modified particle swarm optimization
MR	Multiple regression
MRA	Multiple regression analysis
NDSGA	Non-dominated sorted genetic algorithm
NN	Neural network
NLP	Natural language processing
OD-OS	Oilfield development—offshore structures
OM	Operation & maintenance
PCA	Principal component analysis
PP-SS-PCI	Power plants—substations—power grids
PSO	Particle swarm optimization
RA	Regression analysis
RF	Random forest
RLW	Railways
RBFNN	Radial basis function neural network
RQ	Research question
RRA	Ridge regression analysis
RST	Rough set theory
SBM	Stochastic based model
SBCM	Simulation based cost model
SHAP	Shapley additive explanations
SM	Semantic modelling
SMAA	Sliding moving average approach
SONN	Simplex optimization neural network
SVM	Support vector machine
TUN	Tunneling
UHV	Ultra-high voltage
WPA	Wolf pack algorithm
WTP	Water treatment plants
WWTP	Waste water treatment plants
XGBoosting	Extreme gradient boosting

Introduction

Cost is one of the most important decision-making criteria in the early stages (feasibility) of a construction project design process [1]. The decision to greenlight a construction project hinges on having this information. In the construction industry, cost estimation is the process of predicting the costs required to perform the work within the scope of the project. For decades, accurately estimating construction costs has posed a significant challenge within the construction industry [2]. Precise cost estimation is essential for ensuring the successful completion of a construction project. In today's global competitive landscape, effectively managing construction project costs is key to maintaining competitiveness without compromising on quality standards. [1]. Accurate estimation plays a crucial role in financial management, enabling investors to plan budgets effectively and make informed decisions [2]. Ultimately, the quality of construction management hinges on the precise estimation of construction costs [3].

Various methods exist for estimating construction costs, as outlined below: traditional detailed breakdown cost estimation, cost estimation per activity, simplified breakdown cost estimation, cost estimation based on cost functions, index number estimates, and expert systems [4]. Traditional cost-estimating processes involve a quantity take-off, utilizing blueprints and specifications [1]. Their implementation is a time-consuming and demanding process that requires the commitment of specialized personnel for a significant period. In many cases, these methods have led to significant overruns or underestimations of construction costs as the human factor is involved.

Over the past few decades, technology's rapid and ongoing advancement has influenced the realm of Project-Construction Management. The emergence of Artificial Intelligence (AI) has highlighted new practices and methodologies. Machine Learning (ML) has entered the process of estimating construction costs, leading to new techniques of constructing models for estimating construction costs. The objective of applying ML is to develop models capable of providing swift and precise construction cost estimates while requiring minimal human resources involvement.

Various ML prediction modeling techniques have been introduced for estimating construction costs, including Artificial Neural Networks (ANNs), Regression Analysis (RA), Case-Based Reasoning (CBR), Support Vector Machine (SVM), Fuzzy Logic (FL), Monte Carlo Simulation (MCS), Trees, Forests and Hybrid models. The development of these modeling techniques typically relies on historical data from past projects, and construction experience along with prior knowledge of estimators [2].

The aforementioned ML techniques are applied to construction projects such as Building Structures, Highways/Roadways, Bridges, Water and Wastewater infrastructure and utility projects, Tunneling, Power Plants, Substations, Power Grids, Transmission Lines, Railways, Field Canals, Military Barrack and Facility Projects, Desalination Plants, Sports fields, Urban redevelopment projects, and Underground Gas Storage projects.

Machine learning for construction cost estimation is a rapidly evolving field. Research findings are scattered across various journals, conferences, and technical reports. This study can bridge this gap by offering a comprehensive review and systematic organization of the existing knowledge. In the literature, there is currently no taxonomy study that categorizes in depth the machine learning methods applied and their fields of application in construction cost estimation. Existing review studies lack a detailed and in-depth systematic classification of machine learning techniques and their application areas in construction cost estimation. This hinders researchers from effectively comparing methods and challenges practitioners in choosing the most appropriate technique for their specific cost estimation problems. Additionally, the systematic review studies that exist addressing this subject are limited and include a limited number of relevant articles for analysis. Consequently, a comprehensive systematic review—taxonomy that encompasses a broad range of pertinent articles is currently unavailable. A comprehensive systematic review—taxonomy would provide researchers, practitioners, and decision-makers in the construction industry with a holistic perspective, enabling them to acquire a comprehensive understanding of the diverse range of machine learning techniques applicable to construction cost estimation. A comprehensive systematic review and taxonomy of machine learning techniques for construction cost estimation would contribute to the advancement of knowledge in this field and help to improve the efficiency and effectiveness of construction projects. To address the limitations of existing literature in classifying machine learning techniques for construction cost estimation and their application areas, this study proposes a comprehensive systematic review and an in-depth taxonomy.

The aims of this study are the following:

1. Reviewing the existing ML techniques and their application areas in construction cost estimation.
 2. Generating a novel set of taxonomies by leveraging the findings of the review.
- (a) A classification of the construction areas where the ML techniques are applied.
 - (b) A classification of the ML techniques applied in construction cost estimation.
 - (c) A classification of the correlation between the applied ML techniques and the construction areas.

The proposed set of taxonomies should encompass the following:

Furthermore, this study summarizes the reviewed articles by publication venue (journal or conference), year of publication, and first author. Including an analysis of the number of articles published or presented by journal or conference respectively, provides insights into the leading sources and venues for research on machine learning techniques for construction cost estimation. This also helps in identifying whether there is a growing interest from the scientific community and practitioners in this area. An analysis by year of publication reveals trends and the evolution of the field over time, helping to identify periods of increased activity and make predictions for future growth. Additionally, examining articles by author and their respective contributions highlights the key contributors and influential researchers, facilitating potential collaborations, recognizing the impact of individual authors on the field, and identifying potential challenges in attracting and retaining researchers focused on this specific area.

This systematic review and its accompanying taxonomy of machine learning techniques used in construction cost estimation offer several key benefits and contribute to the body of knowledge in the following ways:

- **Comprehensive overview:** It consolidates and summarizes a wide range of studies, providing a thorough understanding of the current state of the art in machine learning applications for construction cost estimation.
- **Identification of trends:** By reviewing existing literature, the research identifies prevailing trends and emerging techniques, helping researchers and practitioners understand which methods are gaining traction.
- **Classification and organization:** The taxonomy helps in classifying various machine learning techniques and the construction areas in which they are applied into categories, making it easier to navigate the complex landscape of available ML methods in construction cost estimation.
- **Guidance for practitioners:** Practitioners in the construction industry can benefit from a synthesized view of how machine learning can be applied to cost estimation, helping them choose the most appropriate techniques for their specific needs.
- **Identification of gaps and challenges:** The review can uncover gaps in the current research, pointing out areas where further investigation is needed.
- **Facilitation of knowledge transfer:** By summarizing and categorizing existing research, the paper facilitates

knowledge transfer between academia and industry, helping to bridge the gap between theoretical research and practical application.

- **Foundation for future research:** The taxonomy provides a foundation for future research by outlining the existing landscape. Researchers can build upon this foundation to explore new techniques, address identified gaps, and propose novel solutions.
- **Educational resource:** For students and newcomers to the field, the review serves as an educational resource that provides a clear and concise introduction to the topic, helping them get up to speed with the current research landscape.

The rest of the paper is structured as follows: The “Cost estimation techniques and application areas” section offers an overview of pivotal studies on the topic. In the “Research Methodology” section, the systematic literature review’s methodology is outlined, detailing the survey, and establishing the research questions for this article. The “Systematic Review Results—Taxonomies” section presents the outcomes of the review, introducing the novel “AMC” taxonomy. Finally, in “Discussion—Conclusions,” the review’s findings are discussed alongside their implications and future potentials.

Cost estimation techniques and application areas

Machine learning (ML) techniques refer to systems that can learn from experience. They use computational methods to extract information directly from databases, without relying on a predetermined equation. This literature review focuses on machine learning models for construction cost estimation, concerning various types of construction projects. Commonly used machine learning techniques for prediction are applied in building these models.

Numerous studies in the literature concentrate on building structures. Ye Qing and Quan-Feng Wang [5], applied the Backpropagation Neural Network (BPNN) theory to a dataset of 21 building projects in Quanzhou City to develop a model for project cost estimation. Their model exhibited a notable degree of fault tolerance and robust generalization capability. This model effectively computes project costs by concurrently processing data. Test outcomes indicated that the model’s accuracy aligned with the standards for project cost estimation and provided precise design estimates. This study shows that using BPNN offers a valuable tool for reliable project cost estimation.

Some authors used Artificial Neural Networks (ANNs) combined with genetic algorithms for optimized results. Kim et al. [6] employed a Back-Propagation Neural Network (BPNN) model integrated with Genetic Algorithms (GAs) to predict the construction costs of residential buildings.

Genetic Algorithms (GAs) were employed within the BPNN to determine its parameters and enhance the precision of construction cost estimation. The model was trained and evaluated using construction cost data from 530 residential buildings constructed in Korea between 1997 and 2000. This research concluded that a BPNN model integrated with a GA was more accurate and effective in predicting construction costs compared to a BPNN model that relied on trial-and-error adjustments.

Different types of neural networks have been applied in the construction cost estimation of building structures. Jung Seung-Hyun et al. [7] propose that enhanced versions of Deep Neural Network and Deep Belief Network models offer greater effectiveness in achieving precise predictions. They developed a Deep Neural Network (DNN) and a Deep Belief Network (DBN) model specifically tailored for predicting construction costs of smart building schools. This research employed various machine learning models to accurately forecast construction costs even with limited variables and a lack of performance data. The primary aim was to improve the accuracy of construction cost predictions and to facilitate efficient decision-making processes during the planning and design stages of smart education facilities. It’s crucial to acknowledge that higher discrepancies in estimated costs elevate the risks faced by project managers.

Another widely used machine learning technique is Regression Analysis. Alshamrani [8] developed a Multi Regression (MR) model to estimate the conceptual initial costs of both conventional and sustainable college buildings in North America. This model was capable of forecasting the initial cost per square foot for two different structure types: steel and concrete. The development of the model proceeded through three main stages: initial assessment of data quality, model development, and validation. Subsequently, the constructed model underwent successful testing and validation using real-time data.

Fuzzy logic models have been applied in the construction cost estimation of building structures. El-Sawalhi Nabil [9] aimed to create a model to predict parametric cost estimates for building projects in the Gaza Strip utilizing Fuzzy Logic (FL). The study gathered data from 106 real construction projects in the area, focusing on the five most significant factors. The findings indicated that the Fuzzy Model accurately forecasted cost estimates with a satisfactory level of accuracy.

Hybrid models have also been applied in the construction cost estimation of building structures. Koo et al. [10] developed an improved Case-Based Reasoning (CBR) model that integrates the advantages of prediction methodologies like CBR, Multiple Regression Analysis (MRA), and Artificial Neural Networks (ANN), along with an optimization process utilizing a Genetic Algorithm (GA) to forecast construction costs of multifamily housing

projects. The advanced CBR model exhibited an average prediction accuracy of 87.43%. This advanced CBR model is adaptable for expansion. The optimization process utilized in developing the CBR algorithm for various project types could enhance prediction capabilities.

Another efficient hybrid model was proposed by He et al. [11]. A model combining a Fuzzy Neural Network (FNN) with a Modified Particle Swarm Optimizer (MPSO) was utilized for estimating the construction costs of building projects. This approach demonstrated superiority over traditional FNN methods in terms of both generalization capability and automatic parameter optimization. Comparative analyses indicate that this enhanced model outperforms BP (backpropagation) neural networks, PSO-BP, and PSO-FNN models. Case studies showcased the increased accuracy of the developed model in predicting construction project costs. Consequently, this study establishes a framework for informed decision-making in construction project management.

Son Hyojoo et al. [12] presented and validated a hybrid predictive model designed to evaluate cost performance in commercial building projects. This model incorporates 64 variables related to the stages of definition in the pre-project planning phase. Combining Support Vector Regression (SVR) with Principal Component Analysis (PCA), the method approach was analyzed and validated using data from 84 commercial building projects. Additionally, the performance of the PCA–SVR model was compared with four alternative data-mining techniques. Experimental results indicate that the PCA–SVR model proposed accurately forecasts cost performance during the pre-project planning phase, surpassing the efficiency of the other four models.

Khalaf et al. [13] introduced a method for predicting the costs and durations of building structures, utilizing the Particle Swarm Optimization (PSO) technique. The applicability of the PSO in formulating the problem for predicting costs and durations was explored. Sixty projects sourced from completed government projects were employed to develop the proposed models. Findings suggest that PSO models present a practical option for assessing the cost and duration of construction projects, showcasing high accuracy in prediction.

Yiannis Xenidis and Stavrakas [14] utilized two budgeting techniques, the Deterministic method and Monte Carlo Simulation (MCS), in a real case study concerning reinforced concrete works within a building project. A comparison of the outcomes yielded two significant observations: (a) stochastic processes offer more accurate justification for the contingency reserves needed to account for risks in infrastructure project budgets, and (b) stochastic processes are optimal for establishing realistic contingency reserves, resulting in more competitive bids.

Kwon et al. [15] developed a model that integrates Case-Based Reasoning (CBR) with a Genetic Algorithm (GA) to predict maintenance costs for aging residential buildings. An experiment was conducted to assess the model's effectiveness, with the predicted maintenance costs validated using 20 validation datasets. The findings indicated an average similarity of approximately 90% among the retrieved cases. Furthermore, the mean absolute error rate of the cost, adjusted by the Monte Carlo simulation, was approximately 18%, thus affirming the model's reliability.

Numerous authors have focused on construction cost estimation of highways/roadways projects. Aretoulis [16] employed the FANN Tool to construct Artificial Neural Network (ANN) models for forecasting the construction costs of Greek highway projects. The analysis encompassed twenty highway projects constructed in Greece, each providing similar types of data. Utilizing SPSS, correlation analysis was conducted to scrutinize the attributes of each project concerning its actual cost, identifying the most influential variables. Furthermore, the WEKA application, through its attribute selection function, identified the most effective subset of variables. The optimized neural network model attained a mean squared error value of 7.68544E-05 and was based on budgeted cost, lowest awarding bid, technical work cost, and electromechanical work cost.

Tijanić Ksenija et al. [17] developed Artificial Neural Networks (ANNs), including Multilayer Perceptron (MLP), Generalized Regression Neural Network (GRNN), and Radial Basis Function Neural Network (RBFNN), to predict road construction costs. They utilized a database of roads constructed within the territory of the Republic of Croatia for the modeling process. Evaluation of performance among the neural networks revealed that GRNN achieved the highest precision, with a Mean Absolute Percentage Error (MAPE) of 13% and a coefficient of determination of 0.9595. Neural networks have demonstrated potential for application in the preliminary design phase, especially in cases where data for cost analysis are scarce or incomplete. This approach has the potential to provide significantly more precise outcomes and minimizing estimation discrepancies.

Peško Mučenski et al. [18] utilized artificial intelligence to predict costs and durations involved in the reconstruction and/or construction of urban roads. Both Artificial Neural Networks (ANNs) and Support Vector Machines (SVMs) were evaluated and compared. The top-performing SVM exhibited greater accuracy in cost estimation, with a Mean Absolute Percentage Error (MAPE) of 7.06%, in comparison to the most precise ANN, which achieved an accuracy of 25.38%.

Adeli Hojjat and Mingyang Wu [19] introduced a Regularization Neural Network (RNN) and proposed its architecture for estimating construction project costs. The model was specifically utilized to predict the costs of reinforced

concrete pavements. This novel computational model was constructed on a sturdy mathematical basis, ensuring consistently reliable and predictable cost estimations. Moreover, the outcomes derived from the regularization neural network relied solely on the training samples, irrespective of the neural network's structure, learning parameters, or the number of iterations necessary for training the system.

Chou Jui-Sheng [20] presented a parametric estimation method implemented in Texas highway projects, utilizing a set of project characteristics. The author created Generalized Linear Models (GLMs) to forecast initial quantities for activities related to geometry, such as earthwork, pavement, and traffic control, facilitating continuous tracking of project costs. This method of cost breakdown demonstrates the capability to differentiate between quantity uncertainty and price uncertainty in highway construction. The advantage of this method lies in its ability to transition preliminary parametric cost estimates into detailed cost management as project information becomes available during progression. Compared to the conventional practice of employing a gross cost per lane length in the pre-project planning phase, the proposed approach, supported by the developed expert system, provides a more comprehensive foundation and efficient mechanism for monitoring the impacts of changes throughout the project life cycle.

In a bid to improve cost predictions for public roads during early planning stages, Choi et al. [21] proposed a novel conceptual cost prediction model that combines Rough Set Theory (RST), Case-Based Reasoning, and Genetic Algorithms. To identify the most relevant factors influencing construction costs, they used RST alongside in-depth interviews. Then, CBR was applied to forecast road construction costs while acknowledging users' challenges during the conceptual policy planning phase. Finally, a genetic algorithm was used for both the RST and CBR models, leading to an optimized solution. This innovative approach resulted in a more reliable and robust cost prediction model compared to existing methods.

Fragkakis, Nikolaos et al. [22] presented a method for accurate culvert cost prediction in motorways during the early project stages. Their approach leverages a robust database of real-culvert construction projects. By employing linear regression on this data, they can accurately estimate the quantities of materials needed. With these quantities in hand, applying current material unit prices allows for quick and reliable cost estimations. This method's advantage lies in its minimal data requirements, making it ideal for the early project phases and contributing significantly to cost-effective culvert design for motorways to accurate cost prediction for culverts in motorway projects.

There are also studies where the construction cost of bridges was estimated using ML techniques. Karadimos and Anthopoulos [23] employed the Fast Artificial Neural

Network (FANN Tool) to construct Artificial Neural Network (ANN) models aimed at forecasting the construction cost and duration of Greek bridges. They analyzed a dataset comprising 39 bridges constructed in Greece, each accompanied by similar available data. By conducting correlation analysis considering the project's attributes alongside the actual cost and duration, they discerned the most suitable predictive variables. Furthermore, utilizing the attribute selection function in the WEKA application, they identified the most effective subsets of variables. These chosen variables were utilized as input neurons for the neural network models derived from the correlation analysis. The most optimal neural network model, geared towards predicting the actual cost, achieved a mean squared error of $3.84886e-05$, primarily relying on the budgeted cost and the quantity of deck concrete.

Markiz and Jrade [24] presented a model designed to enhance the current methods employed by bridge stakeholders and designers in formulating cost estimates during the conceptual design phase. Their focus included aspects such as project site preparations, substructure, and superstructure of box-girder bridges. They proposed for an integrated approach, combining a decision support system based on Fuzzy Logic (FL) with cost estimation specifically tailored for concrete box-girder bridges. The methodology for developing this model encompasses the integration of an integrated preliminary cost estimation system (IPCES), complex quality functions, and the application of a multi-criteria decision-making (MCDM) strategy.

There are also studies where the construction costs of tunnels were estimated using ML techniques. In a study by Petroutsatou et al. [25] researchers looked at information from 33 twin tunnels totaling 46 km built for the Egnatia Motorway in northern Greece between 1998 and 2004. They aimed to develop two different Neural Networks (NNs)—a Multilayer Feed-Forward Network (MLFFN) and a General Regression Neural Network (GRNN)—that could predict the final construction costs of road tunnels. To see how accurate and reliable these models were, they were compared to actual construction costs. The researchers found that the models they developed were well-suited for their intended purpose, meaning they could potentially provide fairly accurate estimates of costs that would be needed to build road tunnels.

Some authors have focused on construction cost estimation of Water/Wastewater infrastructure and utility projects. Sharma, Jwala et al. [26] have created models to predict the construction, operation, and maintenance (O&M) costs associated with different unit operations and processes within Water Treatment Plants (WTPs). These models were developed utilizing recent historical cost data and can be utilized to produce preliminary cost estimates for key project elements and to assess alternative process configurations

for water treatment projects during their planning stages. Generalized construction cost models were established via regression analysis conducted on the updated construction cost dataset.

Jiang Ru et al. [27] developed a Back Propagation Neural Network (BPNN) to predict the construction costs associated with Wastewater Treatment Plants (WWTPs) in Taiwan. They gathered 26 datasets containing parameters such as design flow rate, influent BOD₅ concentration, and cost details to be used as input variables. The research illustrated that the introduced neural network surpassed linear regression in terms of performance metrics like mean absolute error rate and coefficient of determination. This approach based on neural networks presents a rapid and efficient method for estimating the costs of WWTPs.

Wen Ching-Gung and Chih-Sheng Lee [28] employed Fuzzy Linear Regression Analysis (FLRA) to examine the cost function associated with the construction of wastewater treatment plants, utilizing data from 26 municipal WWTPs in Taiwan. Diverging from traditional regression models where differences between observed and estimated values are typically attributed to measurement errors, this study takes an alternative perspective, interpreting these differences as reflections of the system's parameter fuzziness. Finally, these variances are integrated into a linear function featuring fuzzy parameters. Through the utilization of a linear programming algorithm, this fuzzy linear regression model presents a convenient and practical method for identifying fuzzy structures within an evaluation system.

Gunduz, Murat and Haci Bayram Sahin [29] examined 54 hydroelectric power plants utilizing both Multiple Regression Analysis (MRA) and Artificial Neural Network (ANN) techniques. Subsequently, they formulated two cost prediction models to forecast the construction cost associated with hydroelectric power plants during their early phases. Comparison of validation outcomes revealed that the regression model achieved a prediction accuracy of 9.94%, whereas the neural network model achieved 5.04%. This indicates that the neural network model yielded more precise results.

Hashemi Sanaz Tayefeh et al. [30] developed an Artificial Neural Network (ANN) model integrated with a Genetic Algorithm (GA) to enhance the network architecture for forecasting the construction cost of power plants. The identified optimal network architecture consisted of two hidden layers, achieving an accuracy of 94.71%. Furthermore, they conducted a sensitivity analysis to evaluate the importance of the model's input variables.

Wang, Yuanxiang et al. [31] employed a Genetic Algorithm (GA)-based Support Vector Machine (SVM) and an extreme gradient boosting system for cost prediction of Ultra-high voltage (UHV) power transmission projects. Initially, principal component analysis and expertise from domain experts were combined to identify the key factors of

the subprojects. Subsequently, the genetic algorithm-based support vector machine and extreme gradient boosting system were utilized for cost prediction, respectively. The Shapley value theory from game theory was then employed to determine the weights of combinatorial forecasting, aiming to establish an appropriate combinatorial forecasting model. Finally, an actual UHV project was utilized to demonstrate the prediction accuracy of the proposed combinatorial forecasting model.

Cocodia [32] conducted a comparison between neuro-fuzzy modeling techniques and Multiple Linear Regression Analysis (MLRA) for formulating cost-estimating relationships and forecasting the construction costs of offshore structures at the conceptual phase. The findings were compared against actual project results to assess the accuracy, efficiency, and applicability of both approaches. The comparison demonstrated that in situations where there is a shortage of comprehensive databases or large volumes of historical data, the neuro-fuzzy method surpasses the linear regression models commonly used in the initial phases of offshore construction projects.

Mingyu et al. [33] employed the Microsoft Azure machine learning studio to develop predictive models for oilfield development construction costs, utilizing a machine learning algorithm along with a combination of mathematical and statistical methodologies. Their method integrated three algorithms: a Neural Network (NN), a Boosted Decision Tree (BDT), and a Decision Forest (DF). The findings indicated that the Boosted Decision Tree and Decision Forest algorithms produced permutation feature importance (PFI) rankings with consistent training outcomes. The comparison of the three algorithms revealed that during training, the Decision Forest model showed better performance in terms of average absolute error and squared error loss functions, surpassing both the Neural Network and Boosted Decision Tree models.

Elmousalami, Haytham [34] explores the investigation of 20 AI techniques for conceptual cost modeling, including methods such as fuzzy logic models, artificial neural networks, multiple regression analysis, case-based reasoning, hybrid approaches like genetic fuzzy models, and ensemble methods such as scalable boosting trees (XGBoost) and random forest. The performance evaluation of the applied ML models was conducted using Field Canals Improvement Projects (FCIPs) as an actual case study. Among the 20 AI techniques examined, the findings revealed that XGBoost emerged as the most accurate and suitable method, achieving a mean absolute percentage error of 9.091% and an adjusted R₂ of 0.929.

Gkovedarou Maria and Georgios Aretoulis [35] developed models for forecasting the cost and duration of urban redevelopment projects using Regression Analysis (RA). They also utilized the Fast Artificial Neural Network (FANN

Tool) to create ANNs to predict the project's duration and final cost by incorporating the earthwork volume as an input variable. Both techniques were designed to aid project stakeholders in forecasting the final delivery date and cost of projects, as well as providing early warnings for any deviations from the initial schedule and budget. The findings suggest that neural networks outperform regression analysis models, especially in the context of urban renewal projects.

Research methodology

Driven by our research interests, the gaps identified in the introduction, and the ongoing discussions in academia and industry about using machine learning for construction cost estimation, we conducted this systematic review and built a taxonomy. This research explored how scientists have approached the application of machine learning techniques in this field, aiming to provide insights and understanding of current practices.

Systematic literature reviews are considered secondary-level studies, and the quality of their findings largely depends on the quality of the primary studies they incorporate [36]. Given the substantial volume of relevant articles in the field, it was deemed essential to employ a systematic methodology to initially narrow down the pool of articles based on specific criteria. The methodology adopted adheres to the guidelines introduced by Kitchenham [37, 38]. The criteria for selecting this method relate to its innovative review process, systematic approach, and comparative advantages, including completeness and rigor. These guidelines, oriented towards software engineering, provide the fundamental principles necessary for a comprehensive and systematic literature review, aiding in the selection of qualitative empirical studies and saving time. This method is selected due to its adherence to a uniform protocol, which unfolds in three phases with specific stages. (Outlined in Figure 1) [36]:

- (a) The Planning Phase: The aim is to ascertain the review's contribution and outline the review protocol.
 - (b) The Conducting Phase: The aim is to adhere to the step-by-step review protocol.
 - (c) The Reporting Phase: To deliver an overall presentation and the peer review of the systematic review.
- The subsequent subsections will provide a detailed explanation of each phase involved in the systematic review process.

Phase 1: planning the review

Phase 1 comprises two stages. The initial stage focuses on clarifying the systematic review's necessity and novelty, while the second stage involves drafting the review protocol.

Identification of the need for a systematic review

The "Need" for a systematic review arises from the requirement of researchers and practitioners to summarize all existing information about the application of ML in construction cost estimation in a thorough and unbiased manner, in order to draw more general conclusions than is possible from individual studies or may be undertaken as a prelude to further research activities. While there is a growing body of literature on ML applications in construction cost estimation, there lacks a comprehensive synthesis and categorization of these techniques and their applications. The novelty of the present study is due to the development of taxonomies of the study area based on the results of the systematic review. The taxonomies categorize and develop in-depth the machine learning techniques applied to construction cost estimation, their application field concerning construction projects, and the interconnection between the ML techniques and the application areas.

Development of the review protocol

It is one of the most important stages of the process since it is the stage where the actions that must take place before the Conducting Phase are described and analyzed. The review protocol is refined and upgraded during the systematic review process. Thus, at this stage, the planning concerning the search strategy, the research questions and the selection criteria are identified and discussed.

Phase 2: conducting the review

The Conducting Phase involves executing the actions outlined in the Protocol Development Stage (Phase 1). The stages within this phase are sequential, which may iterate as many activities initiated in the protocol development stage require refinement during the review implementation. [36].

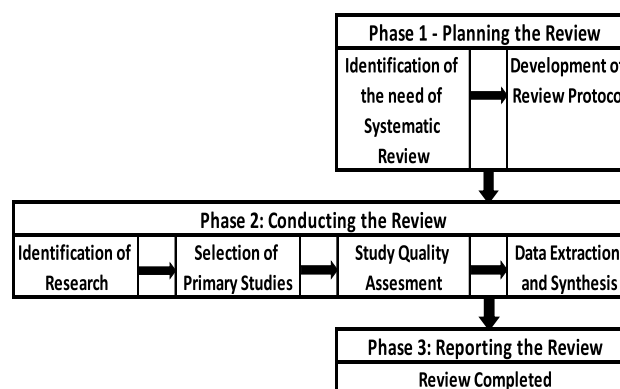


Fig. 1 The phases of the systematic review process [36]

Identification of research

This stage is pivotal in every systematic review as it involves defining the research questions that guide the review's objectives, considering three major perspectives. [36]:

- (a) Population: This refers to the individuals or records relevant to the investigation topic (e.g., studies related to ML in construction cost estimation).
- (b) Interventions: This encompasses alternative methods and approaches related to the topic and/or their comparison (e.g., ML techniques, application areas).
- (c) Outcomes: This includes factors and results that can be used to compare interventions (e.g., algorithms).

By clarifying the review questions, we can determine the type of studies that are most suitable for addressing these questions and establish criteria for including such studies in the review.

The application of ML techniques in construction cost estimation and the studies published in scientific journals or conferences raise many challenges which this review addresses by setting the following research questions:

- RQ1: How many research studies exist that address the application of ML techniques in construction cost estimation?
- RQ2: In which construction areas are ML techniques applied?
- RQ3: Which ML techniques were used for construction cost estimation?
- RQ4: What are the most common ML methods used, and in which construction fields are they most applied?

Selection of primary studies

The following are common methods for searching articles: [39]:

- (a) Manual searching for studies in specific journals and conference proceedings.
- (b) Conducting extensive automated searches in digital libraries.
- (c) The snowball technique (backward or forward).
- (d) A combination of the above methods.

The research strategy of this study relied on a broad automated search in digital sources, spanning from January 1980 to December 2023, with a focus on articles published in journals and conference proceedings. This method has several disadvantages, such as being time-consuming and retrieving many irrelevant articles. However, it remains impartial and exhausts all possible results. Determining

the search terms and selecting the most appropriate digital sources (digital libraries) are essential for implementing the broad automated search method.

Due to their widespread adoption and free access within academic communities, and because their combination has not been utilized in previous similar reviews, the sources used in this study included the following indexing systems and digital libraries: Google Scholar; IEEE Xplore; and ScienceDirect. According to Brophy and Bawden [40], digital libraries (such as ScienceDirect and IEEE Xplore) are preferred for their result quality, while Google Scholar offers extensive accessibility and coverage, with both platforms being accurate. These two search systems were combined to conduct the most comprehensive search for existing research articles relevant to our study.

The search in the aforementioned digital sources involved employing appropriate search strings combined with Boolean operators, adhering to the guidelines established by Spanos and Angelis. [39]. The terms were carefully selected, specifying the search area of construction cost estimation/prediction/forecasting using Artificial Intelligence and Machine Learning in general and also using very popular ML techniques such as Neural Networks (NN), Regression Analysis (RA), Case-Based Reasoning (CBR), Monte Carlo Simulation (MCS), Decision Tree (DT) and Fuzzy Logic (FZ). The search in the digital libraries of Science Direct and IEEE Xplore focused on the titles and the abstracts of the studies, while the search in Google Scholar focused on the titles of the articles. To ensure the identification of a substantial number of articles, a comprehensive set of 30 search terms was utilized. The search terms are presented in Table 1.

The following exclusion selection criteria were implemented:

- (a) Articles conducting studies related to ML techniques and not to construction cost estimation/prediction/forecasting.
- (b) Articles conducting studies referring only to construction cost estimation/prediction/forecasting and not to ML techniques.

Additionally, duplicate articles were removed. The present systematic review does not include all the articles done in this scope but just the ones were found with the defined search terms and in the domain of construction projects.

Study Quality Assessment

The assessment of study quality poses a challenge and is influenced by various factors [37]. In a systematic review, additional criteria are necessary to ensure the high quality

of the included studies. The quality assessment criteria that must be at least partially met for a study to be included in this article are as follows:

- (i) The description of the data, i.e., documentation and description of the methodologies, terms, results, or surveys which presented or cited in the study (i.e., datasets, ML algorithms);
- (ii) The description of the methodology used, (i.e., a comprehensive record of the methodological procedures followed by citing rules and fundamental axioms);
- (iii) The data's availability, i.e., information on access to aforementioned used data (i.e., DOIs, databases, URLs, organizations that provide data);
- (iv) The display of the outcomes (i.e., coherent and comprehensive presentation using tables and graphs).

Table 1 Search terms

A/A	Terms
1	"Cost Estimation" and "Construction"
2	"Cost Prediction" and "Construction"
3	"Cost Forecasting" and "Construction"
4	"Cost Estimation" and "Artificial Intelligence"
5	"Cost Prediction" and "Artificial Intelligence"
6	"Cost Forecasting" and "Artificial Intelligence"
7	"Cost Estimation" and "Machine Learning"
8	"Cost Prediction" and "Machine Learning"
9	"Cost Forecasting" and "Machine Learning"
10	"Cost Estimation" and "Neural Networks"
11	"Cost Prediction" and "Neural Networks"
12	"Cost Forecasting" and "Neural Networks"
13	"Cost Estimation" and "Regression Analysis"
14	"Cost Prediction" and "Regression analysis"
15	"Cost Forecasting" and "Regression analysis"
16	"Cost Estimation" and "Case-Based Reasoning"
17	"Cost Prediction" and "Case-Based Reasoning"
18	"Cost Forecasting" and "Case-Based Reasoning"
19	"Cost Estimation" and "Support Vector Machine"
20	"Cost Prediction" and "Support Vector Machine"
21	"Cost Forecasting" and "Support Vector Machine"
22	"Cost Estimation" and "Monte Carlo"
23	"Cost Prediction" and "Monte Carlo"
24	"Cost Forecasting" and "Monte Carlo"
25	"Cost Estimation" and "Decision Tree"
26	"Cost Prediction" and "Decision Tree"
27	"Cost Forecasting" and "Decision Tree"
28	"Cost Estimation" and "Fuzzy Logic"
29	"Cost Prediction" and "Fuzzy Logic"
30	"Cost Forecasting" and "Fuzzy Logic"

Data Extraction and Synthesis

A final set of studies was extracted following the execution of the Review Protocol steps. The initial number of relevant studies identified was 1,443. After the screening process, the result of the final sample of studies was 219, of which 178 concerned studies were published in scientific journals and 41 concerned conferences' proceedings. Valuable information is extracted from the final sample of articles. The data features extracted from each study, based on our Research Questions, are listed below:

- (a) Publication source, authors, and year of publication—RQ1.
- (b) Article type (Journal/Conference)—RQ1.
- (c) Construction Area—RQ2, RQ4.
- (d) ML Technique—RQ3, RQ4.

Phase 3: reporting the review

During the reporting phase of the review, the systematic review's results are finally presented and assessed. The effectiveness of the review's contribution relies on presenting its results to readers in a clear manner. Hence, it is essential that the completed review is documented, structured properly, and written well to maintain a coherent flow of information. [36].

Systematic review results—taxonomies

This section outlines the survey outcomes.

Data analysis

The initial search in digital libraries using the terms specified above resulted in a total of 1,443 articles. Out of these, 429 articles (29.73%) were identified in Google Scholar, 689 (47.75%) in Science Direct, and 325 (22.52%) in IEEE Xplore. The search in Science Direct and IEEE Xplore with the search terms defined included both the title and abstract of articles, whereas in Google Scholar, the search was limited to just the title of articles, as it is not possible to search the abstracts individually on the Google Scholar platform. This explains why fewer articles were found in Google Scholar compared to what was expected. However, upon verifying the articles found in the other digital libraries, it was noted that all the articles found in Science Direct and IEEE Xplore were also included in Google Scholar.

Taxonomies

The assessed field of this study currently lacks a substantial number of comprehensive review papers, and those that do exist analyze a limited selection of relevant articles. Existing review studies offer valuable insights into construction cost estimation using machine learning. However, they are limited by the absence of an in-depth, taxonomy-based classification system for the machine learning techniques and the specific application areas where they are applied.

The present systematic review facilitates additional analysis of its findings, expanding the systematic approach with a contextualized presentation. This presentation categorizes and organizes the data concerning machine learning techniques applied in construction cost estimation and their respective application domains. Since taxonomies are widely understood and beneficial tools for contextualizing information, they have been selected to highlight knowledge and insights regarding machine learning techniques and their applications in the context of construction cost estimation. The taxonomies were constructed following the procedures outlined by Bennett and Lehman [41]. The novel “A,” “M,” and “C” taxonomies that concern individual taxonomies for Application Area, ML Techniques, and Connection, respectively, return the unified “AMC” taxonomy, with the title given reflecting the first letter of each category.

The “A Taxonomy”: application areas

The systematic review has produced significant findings regarding the application areas of machine learning (ML) techniques in construction cost estimation. Scheme 1 presents the “A Taxonomy” which provides a comprehensive classification of these application areas into basic categories and subcategories. The articles are grouped into subcategories, which fall into the relevant basic categories.

The “A Taxonomy” novelty lies in its extensive and detailed categorization of construction projects where machine learning is applied. It goes beyond simple classification into basic categories such as building projects, highways, railways, bridges, tunneling, etc. Instead, the “A Taxonomy” breaks down these categories into subcategories based on project usage, such as differentiating between the application of ML in residential, commercial, industrial, administrative, educational buildings, etc within the building projects category.

Table 2 shows the number of articles per category and their proportion of the total. The total number of articles related to construction areas is 233, surpassing the total number of 219 articles found in the systematic review.

This is because there are articles where there is application of an ML technique on more than one construction areas and therefore these articles are classified in more than one application area.

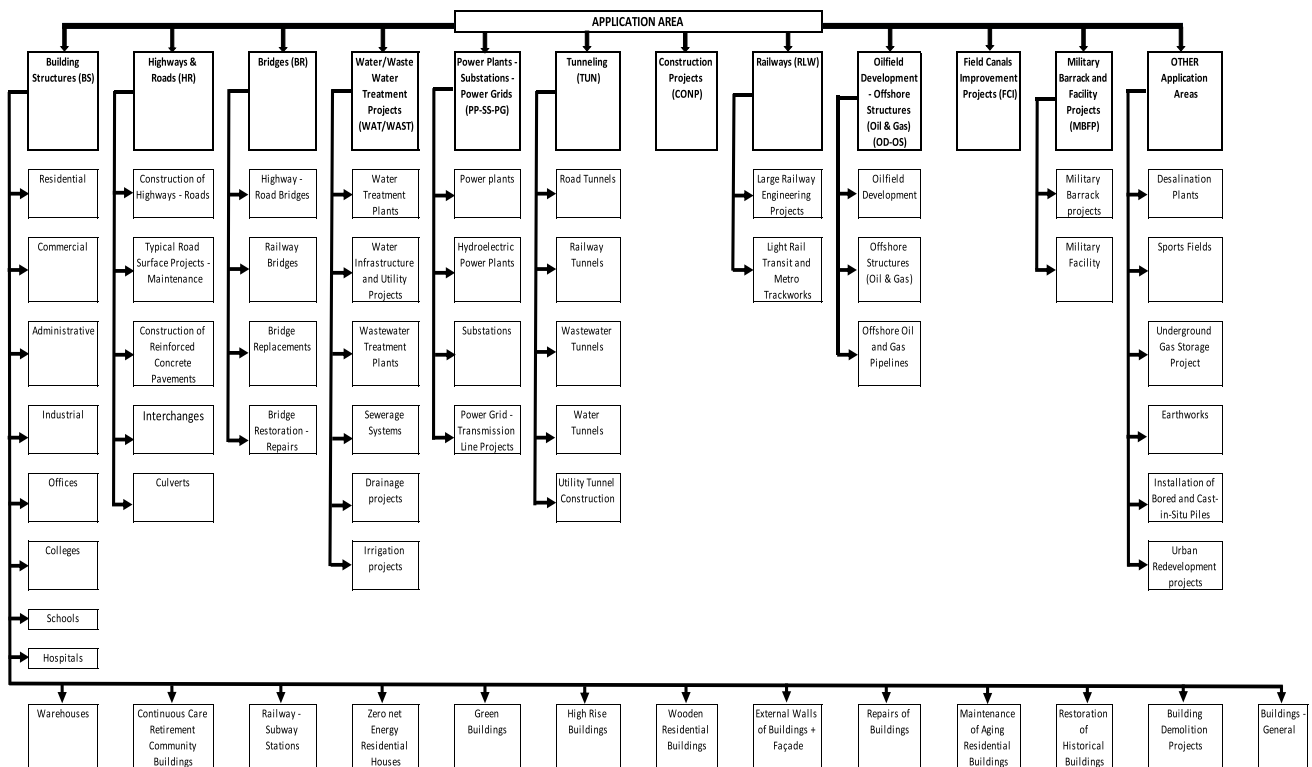
The construction area with the most articles is Building Structures (BS), accounting for over half of the articles in the review (54.51%). This is followed by Highways/Roadways (HR) (14.59%), and then Bridges (5.58%), which have the same percentage of articles as the category of Water/Wastewater infrastructure and utility projects (WAT/WAST) (5.58%). Table 2, also, includes the category of Construction Projects—General (CONP), which encompasses special cases of articles. This category includes articles that refer to construction projects generally, without specifying the construction area [129, 210–213]. The category “OTHER” includes construction areas that appear in no more than one article, except for Earthworks, which corresponds to two articles [133, 221]. Scheme 1 presents the classification of articles into subcategories based on the type of construction projects, such as residential buildings, offices, hospitals, railway bridges, highway bridges, water tunnels, railway tunnels, etc. The articles primarily focus on the complete construction of these projects. However, there are articles that address restoration and repairs [111–113], replacement [184], maintenance [15, 112, 168–170, 171, 177], demolition [126], and redevelopment projects. Additionally, there are articles that predict the construction cost of zero net energy houses [89] and green buildings [90, 147–148], as well as two articles that only address the prediction of the construction cost of specific structural aspects of a building, such as the external walls [105] and facade [154].

The tables in Appendix 1 provide a detailed overview of construction the project-type subcategories and their corresponding literature articles and conference proceedings’ references.

The “M Taxonomy”: ML techniques

The systematic review has produced significant results regarding the application of machine learning techniques in construction cost estimation. In Scheme 2, the classification of these techniques into basic categories and subcategories is depicted in detail under the “M Taxonomy”. The articles have been grouped into their respective subcategories, and each subcategory belongs to a basic category.

The “M Taxonomy” novelty lies in its extensive and detailed categorization of the ML techniques applied in construction cost estimation. It goes beyond simple classification into categories such as ANNs, RA, CBR, SVM, T-F, etc. Instead, the “M Taxonomy” breaks down these categories into subcategories, such as differentiating between the application of Convolutional Neural Networks (CNN),



Scheme 1 The “A” Taxonomy outline

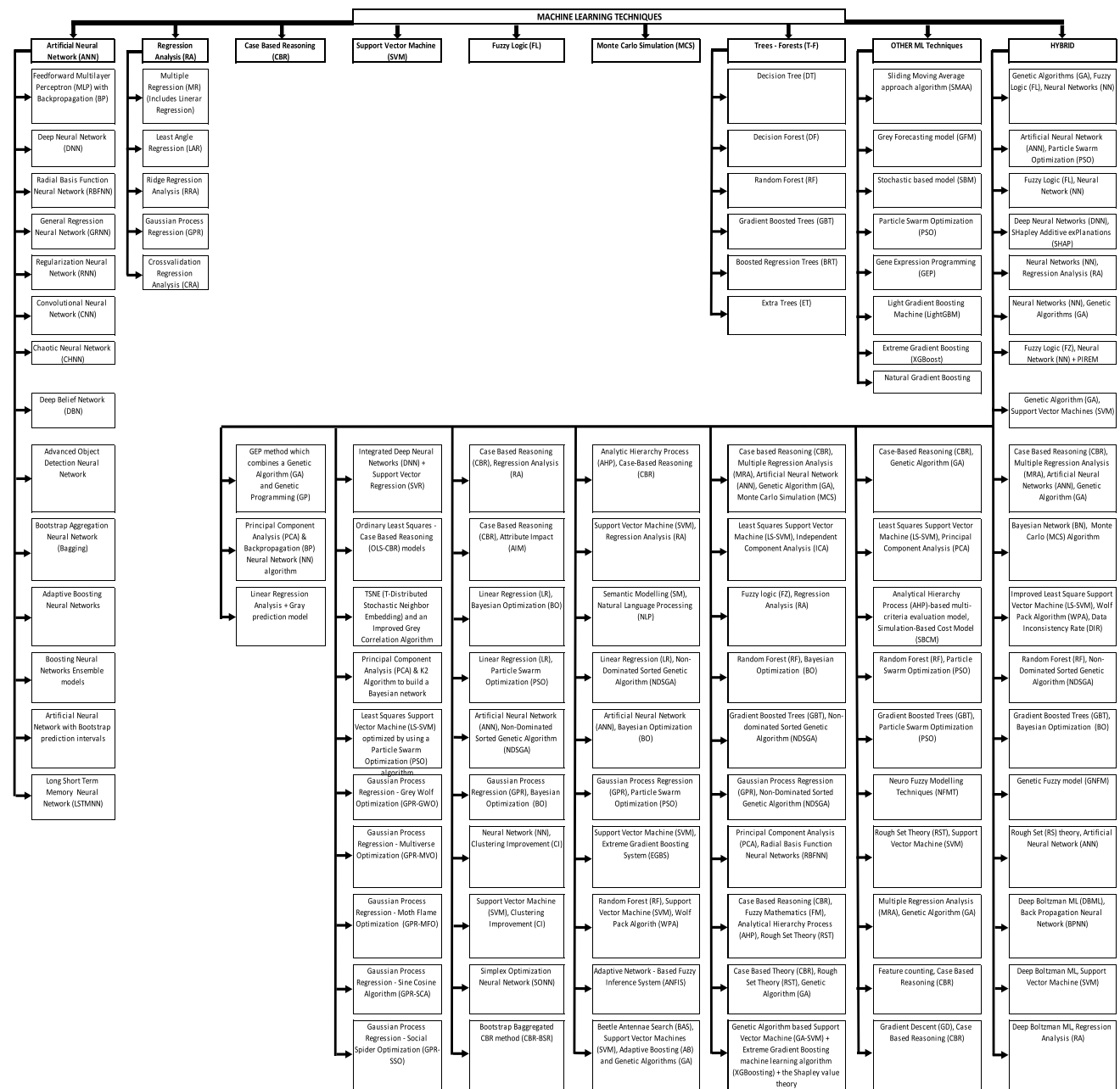
Table 2 Reviewed articles by application area

A/A	Application area	Reviewed Articles	%	Reference No
1	Building structures (BS)	127	54.51	[1–3, 5–15, 42–154]
2	Highways/Roadways (HR)	34	14.59	[16–22, 128, 131, 133, 134, 155–177]
3	Bridges (BR)	13	5.58	[23, 24, 133, 134, 178–186]
4	Water/Wastewater infrastructure & utility projects (WAT/WAST)	13	5.58	[26–28, 127, 128, 133, 134, 187–192]
5	Power Plant—Substation Projects—Power Grid—Transmission Line Projects (PP-SS-PG)	12	5.15	[29–31, 131, 193–200]
6	Tunneling (TUN)	11	4.72	[25, 135, 201–209]
7	Construction projects (general) (CONP)	5	2.15	[129, 210–213]
8	Railways (RLW)	3	1.29	[214–216]
9	Oilfield Development - Offshore Structures (OD-OS)	3	1.29	[32, 33, 217]
10	Field Canals Improvement Projects (FCI)	3	1.29	[34, 146, 218]
11	Military Barrack and Facility projects (MBFP)	2	0.86	[219, 220]
12	Other	7	3.00	[35, 133, 221–225]
	Total	233	100.00	

Chaotic Neural Networks (CHNN), Deep Belief Networks (DBN), Addaptive Boosting Neural Networks, etc within the ANNs category.

Table 3 shows the number of articles found for each basic category and its proportion of the total. The total number of

reviewed articles is 265, which is higher than the number of articles identified in the systematic review (219). This occurs because there are studies in which more than one different machine learning techniques are applied.



Scheme 2 The "M" Taxonomy outline

The most frequently mentioned machine learning technique in the articles is Artificial Neural Networks (ANN), accounting for approximately one-third of the articles (34.72%). The next most prevalent technique is Hybrid models (27.17%), followed by Regression Analysis (RA) (16.23%) and Case Based Reasoning (CBR) (7.55%). Hybrid techniques are defined as those that combine two or more machine learning and/or soft computing methods to achieve higher performance and optimal results. These methods leverage the advantages of multiple methods to attain better performance.

Appendix 2 provides a detailed overview of machine learning subcategories and their corresponding literature articles and conference references.

The "c taxonomy": connection of application areas and methods

Based on the articles reviewed, the Systematic Review highlighted the connection between the type of machine learning used and the construction area. This connection is displayed in "C Taxonomy" and is depicted in Scheme 3a) and

Table 3 Reviewed articles by ML technique

A/A	ML Technique applied	Reviewed articles	%	Reference No
1	Artificial Neural Networks (ANN)	92	34.72	[1, 3, 5, 7, 16–19, 23, 25, 27, 29, 33–35, 42, 43, 45, 47, 49, 50, 53, 54, 56, 57, 59, 62, 63, 66, 68, 69, 71, 72, 76, 80, 83, 86, 89–91, 93, 97, 99–104, 106, 113, 119, 121, 122, 125, 126, 134, 135, 138–141, 143, 145, 147, 149, 154, 158, 160, 161, 163, 165–167, 170, 173, 175–177, 181, 186, 188–192, 204, 211, 213–216, 218, 221]
2	Regression Analysis (RA)	43	16.23	[20, 22, 26, 29, 34, 35, 50, 57, 76, 87, 89, 96, 99, 102, 104, 105, 108, 112, 115, 128, 132, 136, 149, 152, 155, 156, 171, 172, 177, 178, 181, 184, 187, 198, 203–205, 210, 215, 218, 222, 223, 226]
3	Case Based Reasoning (CBR)	20	7.55	[3, 34, 46, 54, 55, 75, 76, 78, 88, 111, 115, 130, 141, 146, 154, 169, 210, 219, 220, 224]
4	Support Vector Machine (SVM)	10	3.77	[18, 34, 45, 58, 60, 99, 123, 149, 163, 176, 181]
5	Fuzzy Logic (FL)	6	2.26	[9, 24, 34, 51, 84, 116]
6	Monte Carlo Simulation (MCS)	6	2.26	[14, 92, 110, 168, 185, 201]
7	Trees—Forests (T-F)	8	3.02	[33, 34, 57, 89, 154, 163, 176, 181]
8	OTHER ML Techniques	8	3.02	[13, 14, 57, 109, 131, 148, 157, 212]
9	HYBRID	72	27.17	[2, 6, 10, 11, 15, 21, 28, 30, 32, 34, 44, 48, 52, 53, 58, 61, 65, 67, 70, 73, 74, 77, 79, 81, 82, 85, 89, 94, 95, 98, 103, 107, 114, 117, 118, 120, 124, 129, 133, 137, 138, 141, 142, 144, 146, 147, 150, 151, 153, 159, 162, 164, 174, 179, 180, 182, 183, 187, 193, 196, 197, 199, 200, 202, 206, 207, 209, 217, 225]
	Total	265	100.00	

Schema 3b). These two Schemes illustrate which machine learning methods have been applied to each construction area. Scheme 3a) begins with the application areas at the top, while Scheme 3b) starts with the machine learning techniques at the top.

The “C Taxonomy” novelty lies in its detailed mapping of the connections between machine learning techniques and their specific application areas in construction. This taxonomy goes beyond a basic listing by visually representing these connections through comprehensive schemes, providing a clear and structured understanding of which ML techniques are applied to various types of construction projects. Furthermore, by offering dual perspectives—starting with application areas in one scheme and ML techniques in another—Taxonomy C enhances the accessibility and usability of the information for researchers and practitioners.

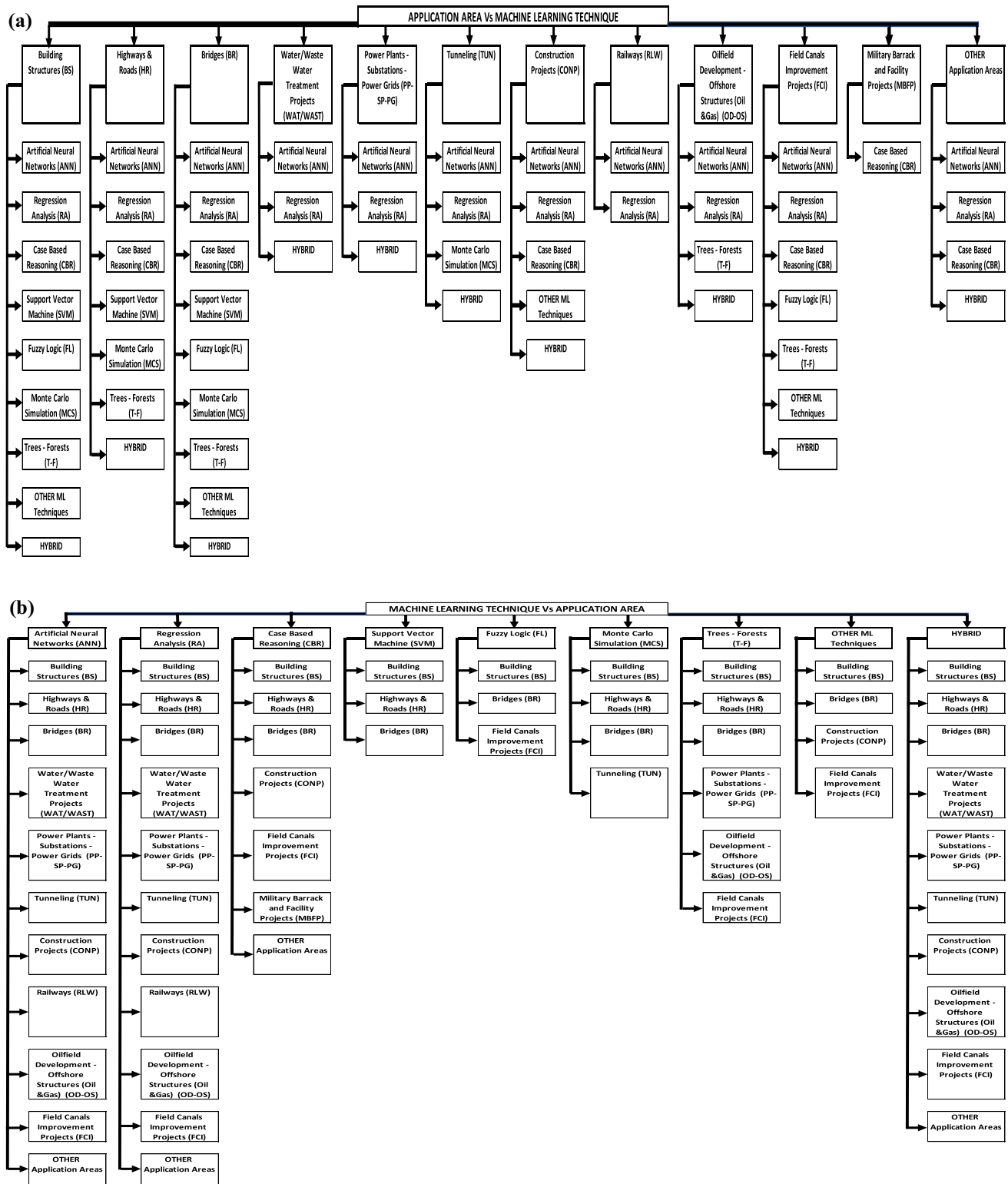
Journals—conferences

Tables 4 and 5 summarize the reviewed articles by journal and conference, respectively, along with their proportion of the total. As demonstrated in Table 4, the journals with the highest number of published articles in the field of construction cost estimation using ML are the Journal of Construction Engineering and Management (13 articles—7.30%), Automation in Construction

(11 articles—6.18%), Expert Systems with Applications (9 articles—5.06%), Building and Environment (6 articles—3.37%), and the International Journal of Project Management (6 articles—3.37%). Table 5 illustrates that all conferences had only one relevant article presented, except for the “4th International Conference on Wireless Communications, Networking, and Mobile Computing” (IEEE, 2008), which had two.

Year of publication

Table 6 summarizes the number of reviewed articles by year of publication (journals and conferences) and their respective proportions of the total. As depicted in the table, the years with the highest number of publications were 2022 and 2011, each with 18 articles (8.22%), followed by 2023, 2021, and 2020, each with 13 articles (5.94%). Figure 2 depicts the distribution of studies over the years. The first study in this field appeared in the literature in 1991. There was a temporary increase in articles in 1998, but the number dropped to zero in 2000. After 2000, there was an almost continuous increase in the number of publications until 2011, which is one of the maximums in the graph. Since then, there has been a decrease in the number of articles found, reaching 8 in 2015. There have been fluctuations in the number of articles up until 2020 and 2021, when the count stabilized at 13. The survey was extended until December 2023, uncovering only 13



Scheme 3 The “C” Taxonomy outline (a). The “C” Taxonomy outline (b)

Table 4 Articles per journal

A/A	Journal	Number of articles	%
1	Journal of Construction Engineering and Management	13	7.30
2	Automation in Construction	11	6.18
3	Expert Systems with Applications	9	5.06
4	Building and Environment	6	3.37
5	International Journal of Project Management	6	3.37
6	Procedia Engineering	5	2.81
7	Journal of Computing in Civil Engineering	4	2.25
8	Advanced Materials Research	3	1.69
9	Buildings	3	1.69
10	Canadian Journal of Civil Engineering	3	1.69
11	Journal of Asian Architecture and Building Engineering	3	1.69
12	Journal of Civil Engineering and Management	3	1.69
13	Tunnelling and Underground Space Technology	3	1.69
14	Advanced Engineering Informatics	2	1.12
15	Applied Sciences	2	1.12
16	Asian Journal of Civil Engineering	2	1.12
17	Computer-Aided Civil and Infrastructure Engineering	2	1.12
18	Construction Management & Economics	2	1.12
19	Engineering, Construction and Architectural Management	2	1.12
20	Journal of Engineering, Design and Technology	2	1.12
21	Journal of Financial Management of Property and Construction	2	1.12
22	Journal of Soft Computing in Civil Engineering	2	1.12
23	Neural Computing and Applications	2	1.12
24	AACE International Transactions	1	0.56
25	Advances in Civil Engineering	1	0.56
26	American Journal of Applied Sciences	1	0.56
27	Applied Mechanics and Materials	1	0.56
28	Applied Soft Computing	1	0.56
29	Arabian Journal of Geosciences	1	0.56
30	Building Research & Information	1	0.56
31	Časopis Građevinar	1	0.56
32	Chemistry and Technology of Fuels and Oils	1	0.56
33	Civil Engineering Journal	1	0.56
34	Cluster Computing	1	0.56
35	Complexity	1	0.56
36	Data in brief	1	0.56
37	Desalination	1	0.56
38	Diyala Journal of Engineering Sciences	1	0.56
39	Engineering Computations	1	0.56
40	Engineering structures	1	0.56
41	Facilities	1	0.56
42	Fuzzy sets and systems	1	0.56
43	HBRC Journal	1	0.56
44	Hindawi Mathematical Problems in Engineering	1	0.56
45	IEEE Access	1	0.56
46	IEEE Transactions on Engineering Management	1	0.56
47	IFAC-PapersOnLine	1	0.56
48	International Journal for Research in Applied Science and Engineering	1	0.56
49	International Journal of Computer Science	1	0.56
50	International Journal of Earth Sciences and Engineering	1	0.56

Table 4 (continued)

A/A	Journal	Number of articles	%
51	International Journal of Emerging Technology and Advanced Engineering	1	0.56
52	International Journal of Engineering Research and Technology	1	0.56
53	International Journal of Information Systems and Social Change (IJSSC)	1	0.56
54	International Journal of Intelligent Computing and Cybernetics	1	0.56
55	International Journal of Intelligent Systems Design and Computing	1	0.56
56	International Journal of New Developments in Engineering and Society	1	0.56
57	International Journal of Operational Research	1	0.56
58	International Journal of Pattern Recognition and Artificial Intelligence	1	0.56
59	International Journal of Project Organisation and Management	1	0.56
60	International Journal of Services Technology and Management	1	0.56
61	International Journal of Structural and Civil Engineering Research	1	0.56
62	International Journal of Structural and Construction Engineering	1	0.56
63	International Journal of Sustainable Built Environment	1	0.56
64	Iranian Journal of Science and Technology, Transactions of Civil Engineering	1	0.56
65	Journal of Architecture Inst. Korea	1	0.56
66	Journal of Artificial Intelligence	1	0.56
67	Journal of Building Construction and Planning Research	1	0.56
68	Journal of Building Engineering	1	0.56
69	Journal of Cleaner Production	1	0.56
70	Journal of Construction Engineering and Project Management	1	0.56
71	Journal of Discrete Mathematical Sciences and Cryptography	1	0.56
72	Journal of Environmental Management	1	0.56
73	Journal of Huaqiao University (Natural Science)	1	0.56
74	Journal of Infrastructure Systems	1	0.56
75	Journal of Management in Engineering	1	0.56
76	Journal of Marine Engineering & Technology	1	0.56
77	Journal of Marine Science and Technology	1	0.56
78	Journal of Natural Gas Science and Engineering	1	0.56
79	Journal of Project Management	1	0.56
80	Journal of the Eastern Asia Society for Transportation Studies	1	0.56
81	Journal of Information Technology in Construction	1	0.56
82	Journal of the Korean Institute of Building Construction	1	0.56
83	Journal of the Korean Institute of Educational Facilities	1	0.56
84	Journal of Rock Mechanics and Geotechnical Engineering	1	0.56
85	Journal of Taibah University for Science	1	0.56
86	Journal of Transportation Systems Engineering and Information Technology	1	0.56
87	Journal of XiamenUniversity (Natural Science)	1	0.56
88	Journal of Xi'an University of Architecture & Technology	1	0.56
89	Korean Journal of Construction Engineering and Management	1	0.56
90	KSCE Journal of Civil Engineering	1	0.56
91	Management	1	0.56
92	Mathematics in Practice and Theory	1	0.56
93	Operational Research	1	0.56
94	Organization, Technology & Management in Construction: An International Journal	1	0.56
95	Procedia Computer Science	1	0.56
96	Procedia—Social and Behavioral Sciences	1	0.56
97	Proceedings of the Institution of Civil Engineers-Smart Infrastructure and Construction	1	0.56
98	Research in Intelligent and Computing in Engineering	1	0.56
99	Revue d'Intelligence Artificielle	1	0.56
100	Structural Safety	1	0.56

Table 4 (continued)

A/A	Journal	Number of articles	%
101	Sustainable construction and building materials	1	0.56
102	Technics Technologies Education Management (TTEM)	1	0.56
103	The Institution of Engineers, Sri Lanka. ENGINEER	1	0.56
104	Tishreen University Journal-Engineering Sciences Series	1	0.56
105	Transportation Research Part A: General	1	0.56
106	Transportation Research Part B: Methodological	1	0.56
107	Tsinghua Science & Technology	1	0.56
108	Tunnel Construction	1	0.56
109	Washington State Transportation Commission	1	0.56
Total		178	100.00

articles for that year, whereas for 2022, the count stands at 18, marking the second-highest point on the graph. It is probable that the count for 2023 is higher, given that the survey concluded in December 2023 and not all relevant articles may have been integrated into the researched digital libraries. It should be noted that during the period 1980–1991 no relevant publications were found.

Author

Table 7 summarizes the number of reviewed articles by author and their portion of the total. Only the first authors of the articles are listed in the table. As shown, the authors with the most publications are Kim, Gwang-Hee (5 articles), Chou, Jui-Sheng (4 articles), and Kim, Sangyong (4 articles). Most of the authors, 167 in total, have written only one article.

Discussion—conclusion

The present study constitutes a systematic review of the studies related to the field of construction cost estimation with the application of ML techniques. Taxonomies related to the ML techniques used in construction cost estimation and their areas of application were developed. The review survey was conducted in Science Direct, IEEE Xplore, and Google Scholar digital libraries and indexing system respectively using specific search terms. The results of the survey are limited to the findings from the above databases and search terms, and the same may not be true if the research was extended to other databases or used different search terms. Additionally, the survey only covers English language articles. However, this study provides an indication of the existing trends.

The following are the novelties and contributions of the present study:

- (a) This is the first time a literature review in this field has been conducted across all three digital libraries (Google Scholar, Science Direct, IEEE Xplore) and covers a period of 44 years, from January 1980 to December 2023.
- (b) For a comprehensive identification of relevant studies, 30 search terms were used, providing more accurate results than any previous review in this field.
- (c) This is the first systematic review in this field with a sample size of 219 studies, providing more accurate results and conclusions.
- (d) The AMC taxonomy, developed in this study, is the first of its kind to analyze the review data in-depth by creating numerous categories and subcategories related to the ML techniques used in construction cost estimation and their application areas.

The results of the systematic review show that the application of ML in construction cost estimation of Building Structures is more intense compared to other construction projects, accounting for 54.51% of the articles. This is mainly due to the following reasons:

- (a) A large amount of capital is invested by private and public investors in building projects, leading to a strong interest in rapid and accurate construction cost estimation which leads to the will to use ML techniques.
- (b) The private sector is more active in building projects than the public sector, making it easier to access available economic and construction data to create ML models.
- (c) Building projects have a wide range of budgets, attracting investors of different economic strata, and this contributes to the further increase of interest in the use of machine learning in construction management.

Table 5 Articles per conference

A/A	Conference	Number of articles	%
1	In 2023 IEEE 12th International Conference on Intelligent Data Acquisition and Advanced Computing Systems: Technology and Applications (IDAACS)	1	2.44
2	In 2023 International Conference on Electronics and Devices, Computational Science (ICEDCS)	1	2.44
3	In 2022 2nd International Signal Processing, Communications and Engineering Management Conference (ISPCEM)	1	2.44
4	In 2022 3rd International Conference on Management Science and Engineering Management (ICMSEM 2022)	1	2.44
5	In 2022 6th Asian Conference on Artificial Intelligence Technology (ACAIT)	1	2.44
6	In 2022 IEEE 6th Information Technology and Mechatronics Engineering Conference (ITOEC)	1	2.44
7	In 2022 International Conference on Intelligent Transportation, Big Data & Smart City (ICITBS)	1	2.44
8	In 2022 Modern Management based on Big Data III	1	2.44
9	In ECITech 2022; The 2022 International Conference on Electrical, Control and Information Technology	1	2.44
10	2022 International Conference on Artificial Intelligence in Everything (AIE)	1	2.44
11	IOP Conference Series: Earth and Environmental Science, 2021	1	2.44
12	In 2021 International Conference on E-Commerce and E-Management (ICECEM). IEEE	1	2.44
13	The 16th IET International Conference on AC and DC Power Transmission (ACDC 2020)	1	2.44
14	In 2019 IEEE 11th International Conference on Humanoid, Nanotechnology, Information Technology, Communication and Control, Environment, and Management (HNICEM)	1	2.44
15	In 2019 IEEE Innovative Smart Grid Technologies-Asia (ISGT Asia)	1	2.44
16	10th International Conference on Measuring Technology and Mechatronics Automation (ICMTMA). IEEE, 2018	1	2.44
17	2017 International Conference on Smart Grid and Electrical Automation (ICSGEA)	1	2.44
18	International Conference on I-SMAC (IoT in Social, Mobile, Analytics and Cloud)(I-SMAC). IEEE, 2017	1	2.44
19	5th Int. Conf. on Civil Engineering and Transportation 2015	1	2.44
20	Proceedings of the International Conference on Electronics, Mechanics, Culture and Medicine (EMCM 2015)	1	2.44
21	In 2015 8th International Symposium on Computational Intelligence and Design (ISCID). IEEE	1	2.44
22	DLSU Research Congress, 2014	1	2.44
23	Procs 28th Annual ARCOM Conference, 3–5 September 2012, Edinburgh, UK	1	2.44
24	Creative Construction Conference 2012 June 30–July 3	1	2.44
25	Second International Conference on Mechanic Automation and Control Engineering. IEEE, 2011	1	2.44
26	In Conference of Korea Institute of Construction Engineering and Management, 2011	1	2.44
27	International Conference on Machine Learning and Cybernetics. IEEE, 2010	1	2.44
28	International Conference on Management and Service Science. IEEE, 2010	1	2.44
29	2010 International Conference on Computer, Mechatronics, Control and Electronic Engineering. IEEE	1	2.44
30	In 2010, The 2nd International Conference on Information Science and Engineering. IEEE	1	2.44
31	International Conference on Computational Intelligence and Natural Computing. IEEE, 2009	1	2.44
32	International Symposium on Automation and Robotics in Construction, 2009	1	2.44
33	International Forum on Computer Science-Technology and Applications. IEEE, 2009	1	2.44
34	Second International Workshop on Knowledge Discovery and Data Mining. IEEE, 2009	1	2.44
35	2008s International Symposium on Intelligent Information Technology Application	1	2.44
36	4th International Conference on Wireless Communications, Networking and Mobile Computing. IEEE, 2008	2	4.88
37	2008 International Symposium on Computational Intelligence and Design. IEEE	1	2.44
38	Third International Conference on Natural Computation (ICNC 2007) IEEE	1	2.44
39	Proceedings of the Korean Institute Of Construction Engineering and Management, 2002	1	2.44
40	14th Annual ARCOM Conference, 1998	1	2.44
	Total	41	100.00

Table 6 Reviewed articles by year of publication

A/A	Year of publication	Number of articles	%
1	2023	13	5.94
2	2022	18	8.22
3	2021	13	5.94
4	2020	13	5.94
5	2019	9	4.11
6	2018	8	3.65
7	2017	12	5.48
8	2016	9	4.11
9	2015	8	3.65
10	2014	11	5.02
11	2013	11	5.02
12	2012	12	5.48
13	2011	18	8.22
14	2010	11	5.02
15	2009	8	3.65
16	2008	10	4.57
17	2007	4	1.83
18	2006	4	1.83
19	2005	6	2.74
20	2004	5	2.28
21	2003	3	1.37
22	2002	3	1.37
23	2001	1	0.46
24	2000	0	0.00
25	1999	1	0.46
26	1998	5	2.28
27	1997	0	0.00
28	1996	1	0.46
29	1995	0	0.00
30	1994	0	0.00
31	1993	1	0.46
32	1992	0	0.00
33	1991	1	0.46
34	1990	0	0.00
	Total	219	100.00

Highways & Road projects follow building projects with a share of 14.59% of the systematic review articles. These projects are mainly public-funded or Public-Private Partnership projects with large investments but fewer investors than building projects. This results in less available financial and construction data and lower interest in developing ML applications for cost estimation than building projects.

The rest of the project categories, accounting for 30.90% of the reviewed articles, have a low percentage of articles for each individual category (<7%), indicating scope for development. The low number of articles for these categories is

due to a lack of interest from researchers, difficulty in finding data, and for some of them due to project complexity.

No relevant articles were found for categories such as Marine projects, Renewable Energy projects, Landfills, and large hydraulic projects like Dams. There is scope for further research and development in these areas.

Table 3 shows that the most used ML technique is Artificial Neural Network (ANN) with 34.72% of the reviewed articles, followed by Hybrid models with 27.17% and Regression models with 16.23%. The large percentage of ANNs is since they work better at predictive analytics because of the hidden layers they have. They use their hidden layers to make predictions more accurate. Linear regression models use only input and output nodes for their predictions. ANNs have also the ability to work with incomplete knowledge, which means that when an ANN is trained, the data may produce output even with incomplete information. The type of ANN that was used the most is the "Feedforward Multilayer Perceptron (MLP) with Backpropagation" which among others is the most widespread type of neural network and it is easy to build.

A large percentage of hybrid models were developed by researchers to integrate different individual prediction models, algorithms, and soft computing. This integration aims to overcome the limitations of individual prediction models, although at the cost of increased complexity in the final model. The goal was to select and combine prediction techniques and algorithms to improve the accuracy of the final prediction models. Hybrid models are advanced-level machine learning techniques. The systematic review shows that the first hybrid models appeared in 1998 [161] and their development has accelerated since 2008. Many of them use Genetic Algorithms to optimize the models. In the present review, the most advanced ML models that appear are hybrids, such as [31], which combines Support Vector Machines and Extreme Gradient Boosting Systems, with the application of Principal Component Analysis and a Genetic Algorithm to estimate the construction cost of an ultra-high voltage (UHV) power transmission project.

The application of regression analysis models was found in 16.23% of the systematic review articles. This high percentage is because regression models, unlike some complex machine learning models, are often relatively easy to understand. The coefficients of the models reveal how much each independent variables influence the dependent variable. This transparency allows you to see not just what the model predicts, but why it makes those predictions. Regarding feature selection the regression analysis can help identify which factors have the most significant impact on the outcome variable. This is useful for selecting the most relevant features for building more complex models and can improve overall model efficiency.

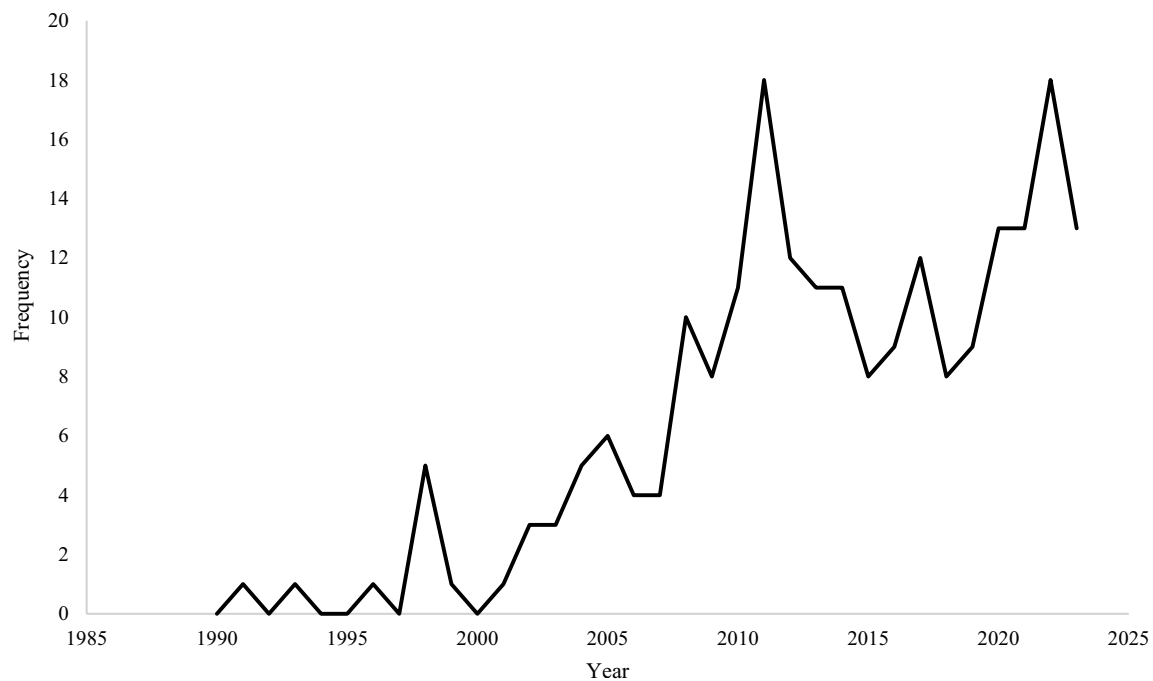


Fig. 2 Distribution of cost estimation studies by year

The remaining ML techniques were found to account for 21.88% of the reviewed articles, however, each individual category accounted for less than 8%. This highlights the potential for further development in these areas. Researchers do not often apply these techniques due to inefficiencies and limitations that may arise in some cases of construction cost estimation.

The AMC Taxonomy has revealed that Artificial Neural Networks (ANNs) and Regression Analysis (RA) techniques were applied to all project categories except for Military Projects (Military Barracks and Facility projects). However, the case of Military projects does not allow us to draw any conclusions about them, given the very small number of articles concerning military projects, only 2. Hybrid techniques were also applied to all categories except for Military Projects and Railways. It can be observed that, in addition to the large number of articles (78.12%) that belong to these three ML techniques categories, there is also a wide application of these techniques in most types of construction projects.

The current systematic review can be highly beneficial for researchers and practitioners, including project and construction managers, contractors, designers, civil, mechanical, and electrical engineers, architects, software developers, and consultants, as it serves as a guide for construction cost estimation using ML techniques and helps in decision making. Scheme 3a and Scheme 3b and also Table 2 and Table 3 allow researchers and practitioners to understand the extent of application of each ML technique in construction cost estimation. By reviewing these schemes and tables, they

can determine which ML techniques have been applied to each project category and to what extent. This information can guide researchers in expanding their current research or developing new research in this area. Additionally, they can identify project categories where specific ML techniques have not been applied, providing opportunities for new studies to be conducted. Practitioners can also use this information to choose appropriate ML techniques, which according to the systematic review and its taxonomies are often used in the project categories of their interest and they have been found to be efficient (e.g., ANN in Building Structures).

The review shows a larger number of studies published in journals (178) compared to the smaller number of studies presented at conferences (41). The 178 journal articles were published in 109 different journals, with only 23 journals having published more than one article, and only 13 having published more than two. The 41 conference studies were presented at 40 different conferences, with only one conference presenting more than one study. The total number (219) of published studies in journals and conferences is small compared to the breadth and importance of the scientific field of construction cost estimation using ML techniques. However, there is a wide distribution of these articles in the literature, as they have been published in many different journals (109) and presented at many different conferences (40). This suggests that despite the limited number of available studies, the scientific press is highly interested in this field. The increasing interest of the scientific press reflects the growing interest of the scientific community

Table 7 Reviewed articles by author

A/A	Author	Number of articles	%
1	Kim, Gwang-Hee	5	2.28
2	Chou, Jui-Sheng	4	1.83
3	Kim, Sangyong	4	1.83
4	El-Sawalhi, Nabil Ibrahim	3	1.37
5	Ji, Sae-Hyun	3	1.37
6	Mahmoodzadeh, Arsalan	3	1.37
7	Ahiaga-Dagbui, Dominic	2	0.91
8	Ahn, Joseph	2	0.91
9	Aretoulis, Giorgos	2	0.91
10	Cheng, Min Yuan	2	0.91
11	Elmousalami, Haytham H	2	0.91
12	Fragkakis, Nikolaos	2	0.91
13	Gunduz, Murat	2	0.91
14	Kim, Byung-Soo	2	0.91
15	Kong, Feng	2	0.91
16	Koo, ChoongWan	2	0.91
17	Mohammed, Hadid Ali	2	0.91
18	Petroutsatou, Kleopatra	2	0.91
19	Sonmez, Rifat	2	0.91
20	Wang, Y.R	2	0.91
21	Wang, Xinzheng	2	0.91
22	Adeli, Hojjat	1	0.46
23	Adumene, Sidum	1	0.46
24	Ahmed, Chro	1	0.46
25	Akanbi, Temitope,	1	0.46
26	Al-Jibouri, Saad HS	1	0.46
27	Alex, Dinu Philip	1	0.46
28	Alias, Melba	1	0.46
29	Alshamrani, Otman Subhi	1	0.46
30	Alshboul, Odey	1	0.46
31	An, Sung-Hoon	1	0.46
32	Antoniou, Fani	1	0.46
33	Arafa, Mohammed	1	0.46
34	Arage, Sphurti S	1	0.46
35	Aziz, Ahmed M. Abdei	1	0.46
36	Bala, Kabir	1	0.46
37	Bai, Yin	1	0.46
38	Barros, Lais B	1	0.46
39	Bayram, Savas	1	0.46
40	Bouabaz, Mohamed	1	0.46
41	Boussabaine, A. H	1	0.46
42	Chakraborty, Debaditya	1	0.46
43	Challa, Ratna Kumari	1	0.46
44	Chandanshive, Viren	1	0.46
45	Chen, Ho-Wen	1	0.46
46	Chen, Yan Chun	1	0.46
47	Cho, Hong Gyu	1	0.46
48	Cho, Hun Hee	1	0.46
49	Cho, N. H	1	0.46
50	Choi, Seokjin	1	0.46
51	Cocodia, E	1	0.46

Table 7 (continued)

A/A	Author	Number of articles	%
52	Concha, Nolan	1	0.46
53	Darko, Amos	1	0.46
54	Dharwadkar, Nagaraj V	1	0.46
55	Doğan, Sevgi Zeynep	1	0.46
56	Du, Zeyan	1	0.46
57	Duan, Xiaochen	1	0.46
58	Elbeltagi, Emad	1	0.46
59	Elhag, T. M. S	1	0.46
60	Elkassas, E.M	1	0.46
61	ElSawy, Ismail	1	0.46
62	Emsley, Margaret W	1	0.46
63	Fan, Miao	1	0.46
64	Feng, Guang Li	1	0.46
65	Feng, Wenhui	1	0.46
66	Gante, Donna Ville	1	0.46
67	Gao, Xiaojun	1	0.46
68	Gkovedarou, Maria	1	0.46
69	Guo, Feng	1	0.46
70	Günaydin, H. Murat	1	0.46
71	Habibi, Farhad	1	0.46
72	Hashemi, Sanaz Tayefeh	1	0.46
73	Hassan, Bassam	1	0.46
74	He, Xin	1	0.46
75	Hegazy, Tarek	1	0.46
76	Hola, Bozena	1	0.46
77	Hong, TaeHoon	1	0.46
78	HongWei, Ma	1	0.46
79	Hyari, Khaled Hesham	1	0.46
80	Inan, Tolga	1	0.46
81	Jafarzadeh, R	1	0.46
82	Ji, Haijun	1	0.46
83	Ji, Zhigang	1	0.46
84	Jiang, Qinghua	1	0.46
85	Jiang, Ru	1	0.46
86	Jin, Dong	1	0.46
87	Jin, RunZhi	1	0.46
88	Jung, Eui Hyunn	1	0.46
89	Jung, Seung Hyun	1	0.46
90	Juszczyk, Michat	1	0.46
91	Karadimos, Panagiotis	1	0.46
92	Khalaf, Tarq Zaed	1	0.46
93	Khan, Mahiuddin	1	0.46
94	Khosrowshahi, F	1	0.46
95	Kim, Ji Myong	1	0.46
96	Kim, Kyong Ju	1	0.46
97	Kovačević, Miljan	1	0.46
98	Kwon, Nahyun	1	0.46
99	Lai, Yu-Ting	1	0.46
100	Li, Hao	1	0.46
101	Li, Heng	1	0.46
102	Li, Qian	1	0.46

Table 7 (continued)

A/A	Author	Number of articles	%
103	Liao, Jing	1	0.46
104	Liu, Jing	1	0.46
105	Liu, Shangke	1	0.46
106	Long, Luong Duc	1	0.46
107	Loutatidou, Savvina	1	0.46
108	Lu, M	1	0.46
109	Lv, Yue	1	0.46
110	Mahalakshmi, G	1	0.46
111	Markiz, Nizar	1	0.46
112	Markovic, Ljubo	1	0.46
113	Marzouk, Mohamed	1	0.46
114	Meharie, Meseret Getnet	1	0.46
115	Mingyu, Shi	1	0.46
116	Mohamed, Basma	1	0.46
117	Pan, Xiao	1	0.46
118	Paraskevopoulou, Chrysothemis	1	0.46
119	Peleskei, Claudius A	1	0.46
120	Peško, Igor	1	0.46
121	Pewdum, Wichan	1	0.46
122	Pham, T. Q. D	1	0.46
123	Qin, Zhongfu	1	0.46
124	Rafiei, M. H	1	0.46
125	Raphael, B	1	0.46
126	Rashid, Ekbal	1	0.46
127	Rostami, Jamal	1	0.46
128	Roxas, C.L	1	0.46
129	Saeidlou, Salman	1	0.46
130	Saito, Mitsuru	1	0.46
131	Senouci, Ahmed	1	0.46
132	Seo, Junwon	1	0.46
133	Setyawati, Bina R	1	0.46
134	Sharma, Jwala R	1	0.46
135	Sharma, Virok	1	0.46
136	Shi, Huawang	1	0.46
137	Shr, Jin-Fang	1	0.46
138	Shrestha, Kabindra K	1	0.46
139	Shutian, Fang	1	0.46
140	Skitmore, R. Martin	1	0.46
141	Sodikov, Jamshid	1	0.46
142	Son, Hyojoo	1	0.46
143	Stoy, Christian	1	0.46
144	Sun, Chun Ling	1	0.46
145	Surenth, S	1	0.46
146	Swei, Omar	1	0.46
147	Tatari, Omer	1	0.46
148	Tawfek, Hany Shoukry	1	0.46
149	Teicholz, Pau	1	0.46
150	Tijanić, Ksenija	1	0.46
151	Ujong, Jesam Abam	1	0.46
152	Uysal, Furkan	1	0.46
153	Vouk, Drazen	1	0.46

Table 7 (continued)

A/A	Author	Number of articles	%
154	Wan, Zhengdong	1	0.46
155	Wang, Haichao	1	0.46
156	Wang, Huey Jiun	1	0.46
157	Wang, Li	1	0.46
158	Wang, Ran	1	0.46
159	Wang, Xing-ji	1	0.46
160	Wang, Yuanxiang	1	0.46
161	Wang, Yingjia	1	0.46
162	Wen, Ching-Gung	1	0.46
163	Williams, Trefor P	1	0.46
164	Wilmot, Chester G	1	0.46
165	Wu, Yunna	1	0.46
166	Xenidis, Yiannis	1	0.46
167	Xiao, Xue	1	0.46
168	XIE, Ying	1	0.46
169	Xin, Zhenyu	1	0.46
170	Xu, Ming	1	0.46
171	Xue, Yang	1	0.46
172	Xue, Xiaojuan	1	0.46
173	Yadav, Richa	1	0.46
174	Yang, Seung-Won	1	0.46
175	Yang, Song-Sen	1	0.46
176	Yau, Nie-Jia	1	0.46
177	Ye, Qing	1	0.46
178	Yi, Lu	1	0.46
179	Yi, Zhouxin	1	0.46
180	Yu, Wen-der	1	0.46
181	Yun, Seokheon	1	0.46
182	Zhao, Hui	1	0.46
183	Zhao, Liang	1	0.46
184	Zheng, Huaihua	1	0.46
185	Zhu, Bao	1	0.46
186	Zima, Krzysztof	1	0.46
187	Žlender, Bojan	1	0.46
188	Zou, Kuancheng	1	0.46
	Total	219	100.00

and practitioners in construction cost estimation using ML techniques.

Figure 2 depicts a steady increase in the number of studies published in the field of construction cost estimation using ML after the year 2000, with a peak in 2011. However, after 2011, there was a decline and stabilization, with the number of articles per year ranging between 8 and 13, except for the year 2022, which experienced a peak of 18 articles. Despite the growing interest from the scientific community and practitioners, the publication of new articles is not keeping pace. This field of research requires a

combination of specialized scientific knowledge and skills in order to be researched effectively, and it is one of the reasons of the limited number of relevant studies. The studies of construction cost estimation using ML need specialized knowledge of civil/mechanical/electrical engineering, construction management, Artificial Intelligence, and Machine Learning, as well as knowledge of databases, depending on the project. The wide range of knowledge required to carry out these studies calls for the involvement of research personnel with the relevant expertise and interests, which necessitates cooperation between them. These requirements

are not easily met. In addition, the absence of organized databases of economic and construction data of projects in many countries hampers the development of new research as it impedes data harvesting for the creation of tables to be used in the development of machine learning models.

The results of the systematic review on the first authors of the articles indicate that only 21 out of 168 authors wrote more than one article, with only 6 authors writing more than two articles. The large number of first authors who only wrote one article suggests that many scientists temporarily worked in the field of construction cost estimation using ML but did not continue their research by producing new studies as first authors. This makes it difficult to cultivate specialized scientists in the field, which in turn hinders the development of the field and the preparation of new studies. To engage more researchers and professionals in this field of research and produce more studies, they need to be motivated. It is proposed to fund more research projects on the construction cost estimation using ML in universities and research centers.

The private sector is also suggested to either fund research projects or develop such research projects in their respective R&D departments. The governments should also increase funding for relevant research projects. Universities and research centers should also join forces and collaborate to produce high-level research in this field.

The systematic review identified 219 articles from January 1980 to December 2023, which is not considered a large quantity for construction cost estimation using ML over these years. The field of research is broad, but the number of related studies is limited, leaving room for further research and the development of new studies. The current review suggests a gap in the application of ML

techniques for construction cost estimation in specific construction areas. According to the review results and the low percentage of identified articles in specific construction areas, there is limited use of ML in construction projects such as bridges, railways, water/wastewater treatment plants, power plants, tunneling, military barracks, and oil-field development projects. Therefore, new studies could be developed to assess the efficiency of ML techniques in these construction areas. Most of the articles found in the systematic review belong to the category of building projects, but even in this category, there is still space for further research and the development of new studies by expanding its subcategories and applying ML to them. Additionally, the proposed new studies could include building ML models for cost estimation in projects such as Marine, Renewable Energy, Landfills, and Dams, where no relevant articles were found. Moreover, it is proposed to use new independent variables to the already developed ML models in order to create new more efficient models, apply ML techniques that have not been widely used, and develop hybrid models using new combinations of methods and algorithms for more accurate forecasting results.

In conclusion, future research in the taxonomy of machine learning techniques for construction cost estimation can benefit significantly from several strategic enhancements. Firstly, expanding the range of search terms can capture a broader spectrum of relevant literature, ensuring a more comprehensive review. Additionally, incorporating searches across more digital libraries, such as Scopus and Web of Science, will diversify the sources and increase the robustness of the findings. Extending the research to include non-English language articles can also provide valuable insights from international studies, which are often underrepresented.

Appendix 1—construction areas subcategories and their references

Building Structures		Highways & Roads		Bridges	
Sub—category	References	Sub—category	References	Sub—category	References
Residential	[1, 3, 6, 10, 70–88, 93–95, 97, 108, 122, 123, 139, 149, 150],	Construction of Highways—Roads	[16–18, 20, 21, 128, 131, 134, 155–167, 169, 172–174, 176]	Highway—Road Bridges	[23, 24, 133, 134, 178–181]
Commercial	[12, 91, 94]	Typical Road Surface Projects—Maintenance	[134, 168–171, 177]	Railway Bridges	[182, 183]
Administrative	[92, 134]	Construction of Reinforced Concrete Pavements	[19, 133, 134, 175]	Bridge Replacements	[184]
Industrial	[93, 108, 110, 127, 131]	Interchanges	[134]	Bridge Restoration—Repairs	[185, 186]
Offices	[95–97, 136]	Culverts	[22]		

Building Structures		Highways & Roads		Bridges	
Sub—category	References	Sub—category	References	Sub—category	References
Colleges	[8, 99, 108, 125]	Water/Wastewater Treatment Projects		Power Plants—Substations—Power Grids	
Schools	[7, 100–104, 134, 151]				
Hospitals	[134]	Sub—category	References	Sub—category	References
Warehouses	[134]	Water Treatment Plants	[26, 188]	Power plants	[30, 131, 196]
Continuous Care Retirement Community Buildings	[106, 107]	Water Infrastructure and Utility Projects	[127, 134, 189–191]	Hydroelectric Power Plants	[29]
Railway—Subway Stations	[109, 115, 152, 153]	Wastewater Treatment Plants	[28, 134, 187]	Substations	[193]
Zero net Energy Residential Houses	[89]	Sewerage Systems	[134, 192]	Power Grid—Transmission Line Projects	[31, 194, 195, 197–200]
Green Buildings	[90, 147, 148]	Drainage projects	[5, 27, 128, 133]		
High Rise Buildings	[98, 124]	Irrigation projects	[134]		
Wooden Residential Buildings	[114]	Tunneling		Railways	
External Walls of Buildings + Façade	[105, 154]				
Repairs of Buildings	[112, 113]	Sub—category	References	Sub—category	References
Maintenance of Aging Residential Buildings	[15]	Road Tunnels	[25, 135, 201–205, 207–209]	Large Railway Engineering Projects	[214, 216]
Restoration of Historical Buildings	[111]	Railway Tunnels	[135, 203, 204]	Light Rail Transit and Metro Trackworks	[215]
Building Demolition Projects	[126]	Wastewater Tunnels	[205]		
Buildings—General (includes articles containing buildings whose use is not mentioned)	[2, 5, 9, 11, 13, 14, 42–69, 116–121, 127–133, 135, 137, 138, 140–146]	Water Tunnels	[205]		
		Utilities Tunnels Construction	[206]		
Oilfield Development—Offshore Structures (Oil & Gas)		Military Barrack and Facility Projects		OTHER Application Areas	
Sub—category	References	Sub—category	References	Sub—category	References
Oilfield Development	[33]	Military Barrack projects	[219]	Desalination Plants	[222]
Offshore Structures (Oil & Gas)	[32]	Military Facility projects	[220]	Sports Fields	[224]
Offshore Oil and Gas Pipelines	[217]			Underground Gas Storage Project	[225]
				Earthworks	[133, 221]
				Installation of Bored and Cast-in-Situ Piles	[223]

Oilfield Development—Offshore Structures (Oil & Gas)		Military Barrack and Facility Projects		OTHER Application Areas	
Sub—category	References	Sub—category	References	Sub—category	References
				Urban Redevelopment projects	[35]

Appendix 2—ML techniques subcategories and their references

Artificial Neural Network (ANN)		Regression Analysis (RA)	
Sub—category	References	Sub—category	References
Feedforward Multilayer Perceptron (MLP) with Backpropagation (BP)	[1, 3, 5, 16–19, 23, 25, 27, 29, 33–35, 42, 43, 47, 49, 57, 62–64, 66, 71, 72, 76, 83, 86, 89, 90, 93, 97, 99–102, 122, 125, 126, 134–136, 138–141, 145, 154, 158, 161, 165–167, 170, 173, 176, 181, 186, 188–190, 204, 211, 215, 221]	Multiple Regression (MR) (Includes Linear Regression)	[20, 22, 26, 29, 34, 35, 50, 57, 76, 87, 89, 96, 99, 102, 104, 105, 112, 115, 128, 132, 149, 152, 155, 156, 172, 177, 178, 184, 187, 198, 203–205, 210, 215, 218, 222, 223, 226]
Deep Neural Network (DNN)	[7, 34, 143, 147, 149]	Least Angle Regression (LAR)	[171]
Radial Basis Function Neural Network (RBFNN)	[17, 18, 53, 56, 63, 90, 119, 121, 190]	Ridge Regression Analysis (RRA)	[156]
General Regression Neural Network (GRNN)	[17, 25, 192, 204]	Gaussian Process Regression (GPR)	[89, 181]
Regularization Neural Network (RNN)	[175]	Crossvalidation Regression Analysis (CRA)	[108]
Convolutional Neural Network (CNN)	[216]		
Chaotic Neural Network (CHNN)	[214]	Trees—Forests (T-F)	
Deep Belief Network (DBN)	[7]	Sub—category	References
Advanced Object Detection Neural Network	[113]	Decision Tree (DT)	[33, 34]
Bootstrap Aggregation Neural Network (Bagging)	[45, 93]	Decision Forest (DF)	[33]
Adaptive Boosting Neural Networks	[45]	Random Forest (RF)	[34, 57, 89, 154, 163, 176, 181]
Boosting Neural Networks Ensemble models	[93]	Gradient Boosted Trees (GBT)	[89]
Artificial Neural Network with Bootstrap prediction intervals	[106]	Boosted Regression Trees (BRT)	[181]
Long Short Term Memory Neural Network (LSTMNN)	[213]	Extra Trees (ET)	[34]
HYBRID (1)		HYBRID (2)	
Sub—category	References	Sub—category	References
Genetic Algorithms (GA), Fuzzy Logic (FL), Neural Networks (NN)	[52]	Gradient Boosted Trees (GBT), Bayesian Optimization (BO)	[89]
Artificial Neural Network (ANN), Particle Swarm Optimization (PSO)	[11, 89, 129, 137, 141]	Genetic Fuzzy model (GNFM)	[34]
Fuzzy Logic (FL), Neural Network (NN)	[44]	Rough Set (RS) theory, Artificial Neural Network (ANN)	[61, 174]
Deep Neural Networks (DNN), SHapley Additive exPlanations (SHAP)	[103]	Deep Boltzman ML (DBML), Back Propagation Neural Network (BPNN)	[58]

HYBRID (1)		HYBRID (2)	
Sub—category	References	Sub—category	References
Neural Networks (NN), Regression Analysis (RA)	[107]	Deep Boltzman ML, Support Vector Machine (SVM)	[58]
Neural Networks (NN), Genetic Algorithms (GA)	[6, 30, 65, 82, 85, 94, 117, 137, 142, 164, 180]	Deep Boltzman ML, Regression Analysis (RA)	[58]
Fuzzy Logic (FZ), Neural Network (NN)+ PIREM	[133]	Case-Based Reasoning (CBR), Genetic Algorithm (GA)	[15, 53, 70, 79, 98, 150, 179, 180, 183, 202]
Genetic Algorithm (GA), Support Vector Machines (SVM)	[73, 193, 206, 207]	Least Squares Support Vector Machine (LS-SVM), Principal Component Analysis (PCA)	[81]
Case Based Reasoning (CBR), Multiple Regression Analysis (MRA), Artificial Neural Networks (ANN), Genetic Algorithm (GA)	[10]	Analytical Hierarchy Process (AHP)-based multi-criteria evaluation model, Simulation-Based Cost Model (SBCM)	[67]
Bayesian Network (BN), Monte Carlo (MCS) Algorithm	[217]	Random Forest (RF), Particle Swarm Optimization (PSO)	[89]
Improved Least Square Support Vector Machine (LS-SVM), Wolf Pack Algorithm (WPA), Data Inconsistency Rate (DIR)	[217]	Gradient Boosted Trees (GBT), Particle Swarm Optimization (PSO)	[89]
Random Forest (RF), Non-Dominated Sorted Genetic Algorithm (NDSGA)	[89]	Neuro Fuzzy Modelling Techniques (NFMT)	[32]
HYBRID (3)		HYBRID (4)	
Sub—category	References	Sub—category	References
Rough Set Theory (RST), Support Vector Machine (SVM)	[118, 120]	Case Based Theory (CBR), Rough Set Theory (RST), Genetic Algorithm (GA)	[21]
Multiple Regression Analysis (MRA), Genetic Algorithm (GA)	[180]	Genetic Algorithm based Support Vector Machine (GA-SVM) + Extreme Gradient Boosting machine learning algorithm (XGBoosting) + the Shapley value theory	[212]
Feature counting, Case Based Reasoning (CBR)	[79]	Analytic Hierarchy Process (AHP), Case-Based Reasoning (CBR)	[156]
Gradient Descent (GD), Case Based Reasoning (CBR)	[79]	Support Vector Machine (SVM), Regression Analysis (RA)	[46]
Case based Reasoning (CBR), Multiple Regression Analysis (MRA), Artificial Neural Network (ANN), Genetic Algorithm (GA), Monte Carlo Simulation (MCS)	[79]	Semantic Modelling (SM), Natural Language Processing (NLP)	[112]
Least Squares Support Vector Machine (LS-SVM), Independent Component Analysis (ICA)	[162]	Linear Regression (LR), Non-Dominated Sorted Genetic Algorithm (NDSGA)	[87]
Fuzzy logic (FZ), Regression Analysis (RA)	[28, 187]	Artificial Neural Network (ANN), Bayesian Optimization (BO)	[2, 87]
Random Forest (RF), Bayesian Optimization (BO)	[89]	Gaussian Process Regression (GPR), Particle Swarm Optimization (PSO)	[87, 206]
Gradient Boosted Trees (GBT), Non-dominated Sorted Genetic Algorithm (NDSGA)	[89]	Support Vector Machine (SVM), Extreme Gradient Boosting System (EGBS)	[31]
Gaussian Process Regression (GPR), Non-Dominated Sorted Genetic Algorithm (NDSGA)	[89]	Random Forest (RF), Support Vector Machine (SVM), Wolf Pack Algorithm (WPA)	[194]
Principal Component Analysis (PCA), Radial Basis Function Neural Networks (RBFNN)	[124]	Adaptive Network—Based Fuzzy Inference System (ANFIS)	[223]
Case Based Reasoning (CBR), Fuzzy Mathematics (FM), Analytical Hierarchy Process (AHP), Rough Set Theory (RST)	[194]	Beetle Antennae Search (BAS), Support Vector Machines (SVM), Adaptive Boosting (AB) and Genetic Algorithms (GA)	[140]

HYBRID (5)		HYBRID (6)	
Sub—category	References	Sub—category	References
Case Based Reasoning (CBR), Regression Analysis (RA)	[95, 150, 182]	TSNE (T-Distributed Stochastic Neighbor Embedding) and an Improved Grey Correlation Algorithm	[151]
Case Based Reasoning (CBR), Attribute Impact (AIM)	[74]	Principal Component Analysis (PCA) & K2 Algorithm to build a Bayesian network	[153]
Linear Regression (LR), Bayesian Optimization (BO)	[89]	Least Squares Support Vector Machine (LS-SVM) optimized by using a Particle Swarm Optimization (PSO) algorithm	[207]
Linear Regression (LR), Particle Swarm Optimization (PSO)	[89]	Gaussian Process Regression—Grey Wolf Optimization (GPR-GWO)	[208]
Artificial Neural Network (ANN), Non-Dominated Sorted Genetic Algorithm (NDSGA)	[89]	Gaussian Process Regression—Multiverse Optimization (GPR-MVO)	[208]
Gaussian Process Regression (GPR), Bayesian Optimization (BO)	[89]	Gaussian Process Regression—Moth Flame Optimization (GPR-MFO)	[208]
Neural Network (NN), Clustering Improvement (CI)	[195]	Gaussian Process Regression—Sine Cosine Algorithm (GPR-SCA)	[208]
Support Vector Machine (SVM), Clustering Improvement (CI)	[195]	Gaussian Process Regression—Social Spider Optimization (GPR-SSO)	[208]
Simplex Optimization Neural Network (SONN)	[164]	GEP method which combines a Genetic Algorithm (GA) and Genetic Programming (GP)	[209]
Bootstrap Baggregated CBR method (CBR-BSR)	[146]	Principal Component Analysis (PCA) & Backpropagation (BP) Neural Network (NN) algorithm	[199]
Integrated Deep Neural Networks (DNN)+ Support Vector Regression (SVR)	[147]	Linear Regression Analysis, Gray prediction model	[200]
Ordinary Least Squares—Case Based Reasoning (OLS-CBR) models	[150]		
OTHER ML Techniques			
Sub—category	References		
Sliding Moving Average approach algorithm (SMAA)	[131]		
Grey Forecasting model (GFM)	[109]		
Stochastic based model (SBM)	[212]		
Particle Swarm Optimization (PSO)	[13]		
Gene Expression Programming (GEP)	[157]		
Light Gradient Boosting Machine (LightGBM)	[57, 148]		
Extreme Gradient Boosting (XGBoost)	[57, 148]		
Natural Gradient Boosting	[57]		

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